Generators of Admissible Quotients: The Structural Constraints of Discovery

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Abstract

A representation learner faced with a quotient space $Q = \mathcal{X}/\!\sim$ must either declare the equivalence relation $\sim$ explicitly (Route A) or learn without structural commitment and risk the three failure modes identified in Paper I: traversal ambiguity, task-map collapse, and the embedding floor. This paper occupies the middle ground. We introduce *generator specifications* — partial structural descriptions that constrain the admissible quotient class without naming $\sim$ explicitly — and show that they impose certifiable architecture constraints.

The main result is a two-step chain: the algebraic closure of a generator set determines the acting group G up to a finite covering ambiguity; the Serre long exact sequence of the principal bundle $G \hookrightarrow \mathcal{X} \to Q$ then computes homotopy invariants of the quotient and a lower bound on $\mathrm{emb}(Q)$. If all admissible quotients require embedding dimension at least m^*, no quotient-faithful encoder into $\mathbb{R}^m$ exists for $m < m^*$, regardless of training. The Hopf fibration is worked in full: a single $U(1)$ generator forces $\mathrm{emb}(Q) \geq 3$, and the bound is sharp.

The theoretical chain is translated into the Generator-Constrained Discovery (GCD) scheme: a four-phase training and certification procedure comprising differentiable optimisation, algebraic closure certification, topology library matching, and architecture correction. The scheme provides counter-pressure against all three Paper I failure modes without requiring $\sim$ to be named in advance.

About this series

A learned representation that achieves perfect task accuracy has not necessarily encoded the equivalence structure of the data-generating process. The series Quotient Faithfulness in Representation Learning develops a unified account of when and why this gap arises and what is required to close it. A representation is quotient-faithful *if it maps semantically equivalent inputs to the same code and faithfully encodes the space of semantically distinct states as a topological structure in the latent space. Three structural manifestations of criterion silence prevent standard training from achieving this: the embedding floor, graded compression, and quotient-respect failure.*

Paper I *[Farkas, 2026a]: Establishes that predictive loss does not certify quotient faithfulness; develops the embedding-space program and the three structural manifestations.*

Paper II *[Farkas, 2026b]: Identifies recovery routes and admissibility sources. Establishes the graded-compression structure and its Kolmogorov complexity foundations.*

Paper III *[Farkas, 2026c]: Derives geometric lower and constructive upper bounds on the latent space dimension required for topology-faithful encoding in the single-quotient setting.*

Paper IV *[Farkas, 2026d]: Extends the dimension-floor programme to region-varying quotient structure via quotient atlases. Develops the given-partition, learned-partition, and emergent-discovery tiers.*

Paper V *[Farkas, 2026e]: Maps the framework against representation-learning traditions and examines fourteen bodies of experimental evidence through the embedding-space lens.*

Paper VI *[Farkas, 2026f] (this paper): Shows that partial structural knowledge — generator specifications — constrains the admissible quotient class and imposes certifiable architecture floors without naming the equivalence relation explicitly. Introduces the Generator-Constrained Discovery (GCD) scheme.*

Paper VII *[Farkas, 2026g]: Empirical validation on benchmark dynamical systems (phase cylinder, Hopf fibration, double pendulum) and an industrial case study (Color Fantasy ship-hotel energy system). Benchmarks against Hamiltonian Neural Networks [Greydanus et al., 2019].*

Contents

1 Introduction

The problem

Papers I and II of this series established that learning a topology-faithful representation of a quotient space $Q = \mathcal{X}/{\sim}$ from finite data requires an *admissibility declaration*: a structural commitment, prior to training, that specifies which equivalence relation $\sim$ is lawful. Without such a declaration, three failure modes are possible regardless of task performance: traversal ambiguity (finite-sample memorisation without quotient structure), task-map collapse (compression of semantically distinct equivalence classes), and the embedding floor (a latent space too small to embed the quotient topology).

Papers 2–4 characterised these failure modes and their conditions. A natural question remains open: *where does the admissibility declaration come from?*

The existing literature offers two answers. The first is explicit declaration: the practitioner specifies $\sim$ in advance by fixing a group G and encoding its action into the architecture — convolutional networks for translation, equivariant networks for rotation, homomorphism encoders for general group actions [Cohen and Welling, 2016, Keurti et al., 2023]. This is Route A of Paper 2, and it works when G is known. The second answer is no declaration: unconstrained representation learning methods such as disentangled VAEs [Locatello et al., 2019] and symmetry discovery methods [Dang-Nhu et al., 2026, Ko et al., 2024] learn structure from data without specifying $\sim$ in advance. They can recover generators but provide no guarantee that the learned representation is quotient-faithful, and do not address the embedding floor.

There is a gap between these positions. In many physical, scientific, and engineering applications, the practitioner does not know $\sim$ exactly, but does know something structural about the data-generating process: that it has conserved quantities, that it obeys a symmetry group of a certain type, that it is generated by a finite set of infinitesimal transformations. This partial knowledge constrains the admissible quotient class without identifying $\sim$ uniquely.

This paper's contribution

This paper formalises that middle ground. The central object is a *generator specification* $G_{\text{spec}} = (S, R_{\text{known}}, \tau, \mathcal{E})$: a finite set of generator symbols, a set of known relations among them, a type tag (Lie or discrete), and an observation space. This is strictly less than naming $\sim$ — the quotient is not declared — but strictly more than no commitment at all.

The main theoretical result is that a generator specification constrains the *admissible quotient class* $\mathcal{Q}(G_{\text{spec}}, \mathcal{X})$: the set of quotient spaces consistent with the specification. The constraint propagates through a two-step chain. The horizontal step (algebraic closure) determines the Lie group G from the generators up to a finite covering ambiguity. The vertical step (the Serre long exact sequence applied to the principal bundle $G \hookrightarrow \mathcal{X} \to Q$) computes homotopy invariants of Q, which in turn determine a lower bound on $\mathrm{emb}(Q)$ via the embedding obstruction results of Paper 3.

The consequence is an *architecture floor*: if all quotients in $\mathcal{Q}(G_{\text{spec}}, \mathcal{X})$ require embedding dimension at least m^*, no quotient-faithful encoder into $\mathbb{R}^m$ exists for $m < m^*$, regardless of training. This is a certifiable constraint derivable from partial structural knowledge alone, without naming $\sim$.

The primary worked example is the Hopf fibration. A single $U(1)$ generator acting freely on S^3 forces $\pi_2(Q) \cong \mathbb{Z}$ via the Serre sequence, identifying $Q \simeq S^2$ and giving $\mathrm{emb}(Q) \geq 3$. The bound is sharp. This example is chosen because the group action is free (no edge cases), the Serre sequence collapses cleanly, and the bound can be verified independently.

The second half of the paper translates the theoretical chain into a differentiable training and certification procedure: the *Generator-Constrained Discovery (GCD) scheme*. The scheme has four phases — differentiable optimisation, algebraic closure certification, topology certification via a library match, and architecture correction — and provides counter-pressure against all three failure modes of Paper I.

Scope and honest limits

Several claims are not made.

The algebraic closure loss $\mathcal{L}_{\text{closure}} = 0$ is a *necessary* condition for the learned matrices to span a Lie algebra, not a sufficient one. Finite-sample closure does not certify quotient faithfulness; this is Open Problem 10.

The topology certification in Phase III is conditional on the quotient family being present in the library. For families outside the library, the floor may be underestimated.

The connection to formal systems in Section 7 establishes that the admissibility framework applies to theorem provers via the lifting constraint language of Spivak [2014], but does not transfer the topological chain to formal type quotients; that requires additional structure and is an open problem.

Relation to Papers 1–4

Paper I established the three failure modes and proved that task sufficiency does not certify quotient faithfulness. Paper 2 characterised admissibility sources and the graded compression. Papers 3–4 established embedding dimension floors via characteristic classes and stochastic escape conditions. The present paper addresses the question that Papers 1–4 leave open: how to derive an admissibility declaration from partial structural knowledge, and what architectural consequences follow.

Organisation

Section 2 introduces generator specifications and the admissible quotient class. Section 3 develops the two-step topological chain. Section 4 contains proofs, regularity conditions, and the two physical regimes. Section 5 handles boundary cases. Sections 6 and 7 generalise to discrete groups and formal systems. Sections 8 and 9 develop the differentiable implementation. Section 10 states open problems.

Summary of Notation

Inherited from Papers 1–4

Symbol	Meaning
$\mathcal{X}$	Observation space
$\sim$	Admissible nuisance relation
$Q = \mathcal{X}/\sim$	Quotient (semantic state) space
$q : \mathcal{X} \to Q$	Quotient map
$E : \mathcal{X} \to Z$	Encoder (representation map)
$Z \cong \mathbb{R}^m$	Latent space of dimension m
$\widetilde{E} : Q \to Z$	Induced quotient representation
E_{label}	Label representation: $E_{\text{label}}(x) = f(x)$
$\text{emb}(Q)$	Topological embedding dimension of Q

New in Paper 6

Symbol	Meaning
$G_{\text{spec}} = (S, R_{\text{known}}, \tau, \mathcal{E})$	Generator specification (Definition 2.1)
$S = \{A_1, \dots, A_k\}$	Finite set of generator symbols
R_{known}	Set of known relations among generators
$\tau \in \{\text{Lie}, \text{discrete}\}$	Type tag
$\mathcal{Q}(G_{\text{spec}}, \mathcal{X})$	Admissible quotient class (Definition 2.3)
G	Acting Lie group determined by S
$\mathfrak{g}$	Lie algebra spanned by generator matrices
$\mathcal{M} = \{M_1, \dots, M_k\}$	Learned generator matrices
$[M_i, M_j]$	Matrix commutator $M_iM_j - M_jM_i$
$P_{\mathcal{M}}$	Orthogonal projection onto $\text{span}(\mathcal{M})$
$T_\alpha(x) = \exp(\sum_i \alpha_i M_i)\, x$	Parametrised group action on $\mathcal{X}$
$\pi_k(Q)$	k-th homotopy group of Q
$\mathcal{L}_{\text{task}}$	Downstream task loss
$\mathcal{L}_{\text{equiv}}$	Equivariance loss (Definition 8.2)
$\mathcal{L}_{\text{closure}}$	Closure loss (Definition 8.5)
$\mathcal{L}_{\text{radial}}$	Radial regulariser (Remark 8.3)
μ, λ, ν	Loss weighting hyperparameters
$\varepsilon_{\text{closure}}$	Closure tolerance threshold (Phase II)
m^*	Embedding dimension floor: $\text{emb}(Q) \geq m^*$

Inherited Results

This paper builds directly on three results from the series. Readers familiar with Papers 1–3 may proceed to Section 2.

Traversal trap [Farkas, 2026a]. For any finite training set $\mathcal{D} \subset \mathcal{X}$, there exists a one-dimensional assignment achieving perfect training loss with no quotient structure. Samplewise fit cannot distinguish a quotient-faithful representation from a one-dimensional traversal.

Compression trap [Farkas, 2026a,b]. The label representation $E_{\text{label}} = f = t \circ q$ is the $\preceq$-minimum among all sufficient representations. If the task map $t : Q \to Y$ is non-injective, E_{label} does not embed Q; it is the unique global minimum of any compression objective preserving prediction. No training criterion alone can certify escape from this minimum.

Dimension trap [Farkas, 2026a,c]. If $m < \text{emb}(Q)$, no continuous map $\widetilde{E} : Q \to \mathbb{R}^m$ is topology-faithful, regardless of training. The embedding dimension floor constrains the encoder output, not any internal layer width. Paper 3 [Farkas, 2026c] derives these floors via characteristic class obstructions and covering dimension bounds.

The present paper addresses the *source* of the admissibility declaration required to escape all three failure modes, and shows that partial structural knowledge — a generator specification — is sufficient to derive certifiable architecture constraints without naming $\sim$ explicitly.

2 Formal Setup: Generator Specifications and Admissible Quotient Classes

Papers 1–4 established that learning a topology-faithful representation of a lawful quotient space $Q = \mathcal{X}/\sim$ from finite data requires an *admissibility declaration*: some structural commitment, prior to or during training, that specifies the equivalence relation $\sim$, its geometry, or at least the class of quotients consistent with the data-generating process. Without such a declaration, three impossibility results apply: the finite-sample traversal, the graded compression, and the dimension trap [Farkas, 2026a].

The four declared admissibility sources identified in Paper 2 — known group actions, causal invariances, augmentation families, and domain knowledge — all share a common structure: the architect names $\sim$ (or a structural sufficient condition for it) before training begins. This is Route A.

The present paper asks the prior question. In many problems of practical interest, the symmetry class itself is unknown: the true group G, if one exists, has not been identified, and no labelled examples of equivalent pairs are available. The architect must choose an architectural bias without knowing which bias is correct. Currently, this choice is made by trial and error over a catalogue of standard symmetry classes (translation, permutation, rotation, ...), with no principled theory of how structural properties of the data-generating process constrain the set of viable choices.

The contribution of this paper is a framework for *generator-constrained discovery*: structural generators of the data-generating process — physical symmetry generators, formal axioms, algebraic consistency requirements — constrain the admissible quotient class without naming $\sim$ explicitly. This is strictly weaker than Route A (which fully declares $\sim$) and strictly stronger than Route A-extension (which applies an architectural bias with no structural justification). In the regime where the hypotheses of the main theorem hold, the generator specification implies topology-level constraints on any admissible quotient, which in turn imply architecture lower bounds via the embedding floor results of Paper 3.

The remainder of this section establishes the three objects on which the theory rests: the generator specification, the admissible quotient class, and the regularity hypotheses under which the main theorem holds.

2.1 Generator Specifications

Definition 2.1 (Generator specification)**.** A *generator specification* for a data-generating process on an observation space $\mathcal{X}$ is a tuple

$$G_{\text{spec}} = (S,\, R_{\text{known}},\, \tau,\, \mathcal{E})$$

where:

1. $S = \{A_1, \ldots, A_k\}$ is a finite set of *generator symbols*, each representing a postulated infinitesimal symmetry of the data-generating process;
2. R_{known} is a (possibly empty) finite set of *known relations* among the generators — algebraic identities or topological constraints that the generating group is known to satisfy;
3. $\tau \in \{\text{Lie, discrete, abstract}\}$ specifies the *generator type*: whether the generators are Lie algebra elements (continuous symmetries), elements of a discrete group presentation, or abstract structural constraints;
4. $\mathcal{E}$ is an (optional) *evidence set* — e.g. persistent homology bars or detected transition symmetries — that supports the postulated generators without constituting a proof.

A generator specification is *complete* if R_{known} contains sufficient relations to determine the global topology of the generating group up to isomorphism. It is *partial* otherwise.

Remark 2.2. Route A corresponds to the special case where G_{spec} is complete and $\tau =$ Lie or $\tau =$ discrete: the architect has declared the full symmetry group. Paper III is concerned primarily with the partial case, where R_{known} is insufficient to determine the global group.

The distinction between specifying only the Lie algebra generators S (Lie algebra data) and specifying the full group including global relations is non-trivial: two non-isomorphic Lie groups can share the same Lie algebra. The canonical example is SO(3) and SU(2): both integrate $\mathfrak{so}(3) \cong \mathfrak{su}(2)$, but $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}/2$ while $\pi_1(\mathrm{SU}(2)) = 0$. A partial specification with $\tau =$ Lie and $R_{\text{known}} = \varnothing$ therefore leaves a *covering ambiguity* that affects the topology of the quotient. This ambiguity is addressed formally in Corollary 2.10.

2.2 The Admissible Quotient Class

Given a generator specification G_{spec} and an observation space $\mathcal{X}$, the *admissible quotient class* is the set of all quotient spaces consistent with G_{spec}.

Definition 2.3 (Admissible quotient class)**.** Let $G_{\text{spec}} = (S, R_{\text{known}}, \tau, \mathcal{E})$ and let $\mathcal{X}$ be the observation space. The *admissible quotient class* is

$$\mathcal{Q}(G_{\text{spec}}, \mathcal{X}) \;=\; \bigl\{ Q \bigm| \exists\, G \text{ with presentation } \langle S \mid R \rangle,\; R \supseteq R_{\text{known}},\; \text{and } Q \simeq \mathcal{X}/G \bigr\}$$

where $\mathcal{X}/G$ denotes the orbit space under a continuous action of G on $\mathcal{X}$, and $\simeq$ denotes homotopy equivalence.

Remark 2.4 (Homotopy equivalence vs. homeomorphism). The admissible quotient class uses homotopy equivalence ($\simeq$) rather than homeomorphism. This choice is deliberate. In the generator-constrained discovery setting, the quotient is not declared; it is discovered progressively during training. Different geometric realisations of the same homotopy type (e.g. a round sphere versus a slightly deformed sphere) are indistinguishable from topological evidence alone. Homotopy equivalence is the natural equivalence for the Serre long exact sequence: the sequence produces homotopy groups, which are homotopy invariants. The embedding dimension bounds derived in Paper 3 from homotopy groups are therefore lower bounds on $\mathrm{emb}(Q)$ for any representative of the homotopy class.

Note that emb is itself a homeomorphism invariant, so homotopy-derived lower bounds may in principle be non-sharp. Sharpening requires homeomorphism-level data (e.g. characteristic classes, intersection forms) beyond what the Serre sequence alone provides. This is an acknowledged limitation discussed further in Section 5.

Remark 2.5. Three features of Definition 2.3 deserve emphasis.

First, $\mathcal{Q}(G_{\mathrm{spec}}, \mathcal{X})$ is a *class* of quotients, not a single quotient. Naming the generators narrows the field of candidates; it does not select a unique candidate. This is the central departure from Route A.

Second, the condition $R \supseteq R_{\mathrm{known}}$ means that every group whose presentation is consistent with the known relations is admitted. A complete specification (R_{known} determines G up to isomorphism) collapses $\mathcal{Q}$ to a single element; a fully empty R_{known} admits all groups generated by $|S|$ generators.

Third, the admissible quotient class depends on both the generator specification and the observation space $\mathcal{X}$. The same generator set acting on different observation spaces produces different quotient classes.

2.3 Regularity Hypotheses

The main theorem requires structural conditions on the group action that ensure the topological machinery (principal bundle structure, Serre long exact sequence) is available. These conditions are stated here explicitly; the consequences of relaxing each are discussed in Section 5.

Definition 2.6 (Regular generator-quotient pair)**.** A pair $(G_{\mathrm{spec}}, \mathcal{X})$ is *regular* if the following hold:

- **(H1) Lie type.** $\tau = \mathrm{Lie}$ and G is a connected compact Lie group.
- **(H2) Free proper action.** G acts freely and properly on $\mathcal{X}$: the stabiliser of every point is trivial and the action map is proper.
- **(H3) Contractible observation space.** $\mathcal{X}$ is contractible: $\pi_k(\mathcal{X}) = 0$ for all $k \geq 1$.

Remark 2.7 (Scope of H1–H3). Hypothesis H1 excludes non-compact groups (e.g. the Lorentz group, scaling symmetries) and discrete symmetry groups. The discrete case is treated separately in Section 6.

Hypothesis H2 is the key structural condition. It ensures that the orbit projection $\mathcal{X} \to \mathcal{X}/G$ is a principal G-bundle, not merely a quotient map. Free actions are violated when G has fixed points (e.g. $\mathrm{SO}(3)$ acting on $\mathbb{R}^3$ fixes the origin). Proper actions are violated for non-compact groups acting on compact spaces. When H2 is violated, the orbit space is stratified and the global fibration structure breaks down; see Remark 2.13.

Hypothesis H3 is satisfied by the most common observation spaces in machine learning: $\mathcal{X} = \mathbb{R}^n$ is contractible, as are open convex subsets of $\mathbb{R}^n$, and any star-shaped domain. It is *not* satisfied by sphere-valued observations (S^n for $n \geq 1$), tori, or compact manifolds. The case of non-contractible $\mathcal{X}$ requires retaining the full long exact sequence and constitutes a separate extension; see Remark 2.13 and Section 5.

2.4 Main Results

We now state the three results that form the theoretical core of the paper. Proofs are assembled in Section 4.

Proposition 2.8 (Known-quotient baseline)**.** *Suppose the quotient Q is fully declared (Route A): the architect specifies $\sim$ explicitly and $Q = \mathcal{X}/{\sim}$ is known. Then by Paper 3* [Farkas, 2026c]*, the embedding floor*

$$\mathrm{emb}(Q) \;\geq\; F\big(\pi_1(Q),\, \pi_2(Q),\, \mathrm{dm}(Q)\big)$$

applies directly. Any encoder $E\colon \mathcal{X} \to \mathbb{R}^m$ with $m < \mathrm{emb}(Q)$ is in the embedding floor regardless of the training procedure.

Theorem 2.9 below extends this baseline to the generator-constrained setting: when Q is not declared but generators G_{spec} are known, the identical bound applies to every $Q \in \mathcal{Q}(G_{\mathrm{spec}}, \mathcal{X})$.

Theorem 2.9 (Regular generator-to-quotient constraint)**.** *Let $(G_{\mathrm{spec}}, \mathcal{X})$ be a regular pair in the sense of Definition 2.6, and let $Q = \mathcal{X}/G$ for any group G with presentation $\langle S \mid R\rangle$ consistent with G_{spec}. Then:*

- (i) *The orbit projection $q\colon \mathcal{X} \to Q$ is a principal G-bundle.*
- (ii) *Since $\mathcal{X}$ is contractible, the Serre long exact sequence of homotopy groups collapses to isomorphisms:*
$$\pi_k(Q) \;\cong\; \pi_{k-1}(G) \quad \textit{for all } k \geq 1.$$
- (iii) *The homotopy groups $\pi_k(Q)$ are therefore determined by the topology of G, which is constrained (up to covering ambiguity, see Corollary 2.10) by the generator specification G_{spec}.*
- (iv) *By the embedding floor results of Paper 3* [Farkas, 2026c]*, there exists an explicitly computable function F such that*
$$\mathrm{emb}(Q) \;\geq\; F\big(\pi_1(G),\, \pi_2(G),\, \mathrm{dm}(Q)\big)$$
for every $Q \in \mathcal{Q}(G_{\mathrm{spec}}, \mathcal{X})$.

Consequently, the generator specification G_{spec} alone — without declaring $\sim$ — determines a lower bound on the latent dimension required for any quotient-faithful encoder $E\colon \mathcal{X} \to \mathbb{R}^m$.

Corollary 2.10 (Covering ambiguity limits Lie algebra data)**.** *Under the hypotheses of Theorem 2.9, the Lie algebra generators S alone (without global relations R_{known}) determine $\pi_k(G)$ for $k \geq 2$ exactly, but determine $\pi_1(G)$ only up to the finite group of deck transformations of the universal covering $\tilde{G} \to G$. Therefore a partial specification with $R_{\mathrm{known}} = \varnothing$ yields a* family *of lower bounds on $\mathrm{emb}(Q)$, indexed by the admissible covering choices. The sharpest bound corresponds to the simply connected integration $\tilde{G}$; the weakest to the quotient with the largest $|\pi_1(G)|$.*

Proposition 2.11 (Architecture obstruction)**.** *Under the hypotheses of Theorem 2.9, if*

$$\min_{Q \in \mathcal{Q}(G_{\mathrm{spec}}, \mathcal{X})} \mathrm{emb}(Q) \;>\; m,$$

then no continuous encoder $E\colon \mathcal{X} \to \mathbb{R}^m$ is quotient-faithful for any $Q \in \mathcal{Q}(G_{\mathrm{spec}}, \mathcal{X})$. In particular, setting the latent dimension to m guarantees the embedding floor regardless of the training objective or the amount of training data.

Remark 2.12 (Relation to the three failure modes). Proposition 2.11 is a direct extension of the embedding floor (Paper I, Theorem 3): the dimension floor now applies to the entire admissible quotient class rather than to a single declared quotient. The finite-sample traversal and graded compression are not addressed by Theorem 2.9 alone, which is a purely topological statement; the mechanisms by which a training scheme can resist those traps are discussed in Section 9.

Crucially, Theorem 2.9 does not assert that a generator-based training scheme will find a quotient-faithful encoder. It asserts that if the latent dimension is below the floor, no such encoder exists. The theorem provides a necessary condition on architecture, not a sufficient condition on learning.

Remark 2.13 (Singular and non-contractible extensions). When H2 is violated (non-free action), the orbit space is a stratified space: the principal stratum (free orbits) retains the bundle structure and the homotopy isomorphisms of Theorem 2.9, but singular strata (orbits with non-trivial stabilisers) require separate treatment via orbifold cohomology or equivariant homotopy theory. The bounds of Proposition 2.11 apply on the principal stratum and serve as lower bounds for the full space.

When H3 is violated (non-contractible $\mathcal{X}$), part (ii) of Theorem 2.9 no longer holds; the Serre sequence does not simplify to isomorphisms, and $\pi_*(Q)$ depends on both $\pi_*(G)$ and $\pi_*(\mathcal{X})$. The primary worked example of Section 3.3 (the Hopf fibration) operates in this non-contractible regime and demonstrates that the full long exact sequence is still usable and yields sharp results.

Remark 2.14 (Empirical instance: altermagnet discovery). The generator-constrained discovery framework has a concrete recent instance in condensed-matter physics. Altermagnets — a third branch of magnetic order, distinct from both ferromagnetism and antiferromagnetism — were identified by narrowing the admissible class of magnetic phases via a specific generator specification: rotation-plus-time-reversal-translation symmetry groups acting on spin configurations [Hassani-Vasmejani et al., 2026]. The generator constraint narrowed the admissible quotient class sufficiently that new structure (anisotropic spin splitting with d-wave, g-wave, and i-wave order parameters) fell out of the classification. This is a current worked case of the scheme formalised in Theorem 2.9: generator specifications alone, without full declaration of $\sim$, determined a structure-preserving quotient class from which genuine scientific discovery followed.

3 From Generators to Quotient Topology

The main theorem constrains the admissible quotient class via a two-step mechanism. The first step is algebraic: partial generator knowledge can be extended to a complete Lie algebra by enforcing closure under the Lie bracket. The second step is topological: once the Lie algebra (and hence the group topology, up to covering ambiguity) is determined, the Serre long exact sequence maps generator data to homotopy groups of the quotient, and the embedding floors of Paper 3 give architecture lower bounds.

We call these the *horizontal step* (algebraic closure, expanding partial knowledge within the generator level) and the *vertical step* (topological bounding, descending from generators to quotient topology to latent dimension).

3.1 Algebraic Closure: The Horizontal Step

A Lie algebra $\mathfrak{g}$ is, by definition, a vector space closed under the Lie bracket $[\cdot\,,\,\cdot]\colon \mathfrak{g}\times\mathfrak{g} \to \mathfrak{g}$. If a learning system discovers a set of generators $S = \{A_1, \ldots, A_k\}$ that does *not* close under the bracket — i.e. there exist $A_i, A_j \in S$ such that $[A_i, A_j] \notin \mathrm{span}(S)$ — then S is a *partial* generator set: it generates a subalgebra of the true symmetry algebra but not the full algebra.

Definition 3.1 (Lie algebraic closure)**.** Let $S = \{A_1, \dots, A_k\}$ be a finite set of elements in some Lie algebra. The *Lie algebraic closure* of S, denoted $\overline{S}$, is the smallest Lie subalgebra containing S; equivalently, the intersection of all subalgebras containing S. Concretely, $\overline{S}$ is computed by the following completion procedure:

1. Set $B_0 = \mathrm{span}(S)$.
2. At step n, compute $[A_i, A_j]$ for all $A_i, A_j \in B_n$. Let $r_{ij} = [A_i, A_j] - \mathbf{P}_{B_n}([A_i, A_j])$, the component of $[A_i, A_j]$ orthogonal to B_n.
3. If any $r_{ij} \neq 0$, set $B_{n+1} = B_n \oplus \mathrm{span}\{r_{ij}\}$ and repeat; otherwise $\overline{S} = B_n$.

The process terminates because the dimension of B_n is bounded by $\dim \mathfrak{g}$, which is finite for compact Lie groups.

Remark 3.2 (Algebraic closure vs. Route A). Algebraic closure is a strictly weaker commitment than Route A. Route A declares the full group G (equivalently, the full algebra with all its relations). Algebraic closure declares only the bracket structure: it identifies which algebra $\overline{S}$ is generated by the partial set S, but leaves open the covering ambiguity of Corollary 2.10 (whether the true group is G or a covering of G with the same algebra).

In the SO(3)/SU(2) example: knowing the generators $\{L_x, L_y\}$ of $\mathfrak{so}(3)$ is sufficient for the bracket $[L_x, L_y] = L_z$ to close the algebra immediately to the full $\mathfrak{so}(3) \cong \mathfrak{su}(2)$. The algebraic closure is $\overline{\{L_x, L_y\}} = \mathfrak{so}(3)$. But whether the true group is SO(3) or SU(2) is not determined by the algebra alone; it requires global data (e.g. whether large-angle transitions satisfy $\exp(2\pi L_i) = I$). This is the covering ambiguity; it does not prevent the algebraic step from being taken, but it does mean that the topological step (Section 3.2) will produce a *family* of bounds rather than a single sharp bound.

Remark 3.3 (Partial knowledge that does not close trivially). Not all partial generator sets close immediately. Genuinely partial knowledge arises when the discovered generators generate a *proper subalgebra.* The canonical example relevant to this paper is a single generator A corresponding to a $U(1)$ phase symmetry in a system whose true symmetry is SU(2). The closure $\overline{\{A\}} = \mathfrak{u}(1)$ (one-dimensional, abelian) is a proper subalgebra of $\mathfrak{su}(2)$. Acting with this single generator on the observation space $\mathcal{X} = S^3$ produces the Hopf fibration, worked out in detail in Section 3.3.

The distinction between "knowing two generators that already close the full algebra" (like L_x, L_y for $\mathfrak{so}(3)$) and "knowing a single generator that generates a proper subalgebra" (like A for $\mathfrak{u}(1) \subset \mathfrak{su}(2)$) is important: the former is not genuinely partial knowledge in the algebraic sense, while the latter is. The paper's framework is primarily concerned with the latter.

In the learning setting, the generator matrices $\mathcal{M} = \{M_1, \dots, M_K\}$ are not given analytically; they are learned from data. The algebraic closure condition then becomes a measurable quantity: the commutator residual

$$\mathcal{L}_{\mathrm{closure}}(\mathcal{M}) \;=\; \sum_{1 \le i < j \le K} \big\| [M_i, M_j] \;-\; \mathbf{P}_{\mathcal{M}}\big([M_i, M_j]\big) \big\|_F^2, \tag{1}$$

where $\mathbf{P}_{\mathcal{M}}$ is orthogonal projection onto $\mathrm{span}(M_1, \dots, M_K)$ and $\|\cdot\|_F$ is the Frobenius norm.

Proposition 3.4 (Closure residual as a diagnostic)**.** $\mathcal{L}_{\mathrm{closure}}(\mathcal{M}) = 0$ *if and only if*

$\mathrm{span}(M_1, \ldots, M_K)$ *is closed under the matrix commutator. If* $\mathcal{L}_{\mathrm{closure}}(\mathcal{M}) > 0$*, then the residual direction*

$$M_{K+1} \;\propto\; \sum_{i<j} \big([M_i, M_j] - \mathbf{P}_{\mathcal{M}}([M_i, M_j])\big)$$

is a natural candidate for an additional generator that extends the span toward closure.

Proof. The first statement is immediate from the definition of the projection $\mathbf{P}_{\mathcal{M}}$. For the second, the residual is the component of the commutators not explained by the current span; adding it to the basis reduces $\mathcal{L}_{\mathrm{closure}}$ to zero in one step if the residual is itself in the closure, or initiates the next iteration of Definition 3.1 otherwise. □

Remark 3.5 (What $\mathcal{L}_{\mathrm{closure}} = 0$ does and does not imply). A vanishing closure residual implies that the learned span is approximately closed under the matrix commutator. It does *not* imply:

- that the structure constants are stable or uniquely identifiable from finite data,
- that the algebra is semisimple or compact,
- that the learned matrices integrate to a specific global group G (the covering ambiguity of Corollary 2.10 remains),
- that the encoder E_θ maps the orbits of G faithfully into the latent space (the sufficiency gap; see Open Problem 10).

These limitations are inherited from the general setting; they do not invalidate the use of $\mathcal{L}_{\mathrm{closure}}$ as a structural diagnostic.

3.2 Topological Bounding: The Vertical Step

Once the Lie algebra $\mathfrak{g} = \overline{S}$ has been determined (or approximated) by the algebraic closure step, the Serre long exact sequence of the principal bundle $G \to \mathcal{X} \to Q$ maps $\pi_*(G)$ to $\pi_*(Q)$. Under the regularity hypotheses of Definition 2.6, with $\mathcal{X}$ contractible, the sequence collapses to the isomorphisms $\pi_k(Q) \cong \pi_{k-1}(G)$ of Theorem 2.9(ii).

Remark 3.6 (The sequence in the non-contractible case). When $\mathcal{X}$ is not contractible (H3 violated), the full sequence

$$\cdots \to \pi_k(G) \to \pi_k(\mathcal{X}) \to \pi_k(Q) \to \pi_{k-1}(G) \to \cdots$$

must be used. This requires $\pi_*(\mathcal{X})$ as additional input. The Hopf example of Section 3.3 is of this type ($\mathcal{X} = S^3$, $\pi_3(S^3) = \mathbb{Z}$) and demonstrates that the sequence remains tractable and yields sharp results even without contractibility.

The vertical step is summarised schematically as follows:

$$G_{\mathrm{spec}} \xrightarrow{\text{closure}} \mathfrak{g} = \overline{S} \xrightarrow{\text{integration}} G \text{ (up to covering)} \xrightarrow{\text{Serre}} \pi_*(Q) \xrightarrow{\text{Paper 3}} \mathrm{emb}(Q) \geq F(\pi_*(Q)).$$

Each arrow is a theorem, not a heuristic. The integration arrow carries the covering ambiguity; every other arrow is canonical.

3.3 Primary Worked Example: The Hopf Fibration

We now run the full chain from a single generator to an architecture lower bound. This example operates in the non-contractible regime (H3 violated) and therefore demonstrates the full Serre sequence rather than the simplified isomorphism.

Setup. Let $\mathcal{X} = S^3$, the unit 3-sphere. This models, for example, the state space of a quantum two-level system (where states are unit complex vectors in $\mathbb{C}^2 \cong \mathbb{R}^4$ up to global phase) or the space of unit quaternions representing 3D orientations.

Partial generator knowledge. The learning system has discovered a single continuous conserved quantity corresponding to a phase rotation. The generator specification is $G_{\text{spec}} = (\{A\}, \varnothing, \text{Lie}, \mathcal{E})$ for a single Lie algebra element A generating the Lie algebra $\mathfrak{u}(1)$. Algebraic closure is immediate: $\overline{\{A\}} = \mathfrak{u}(1)$ (one-dimensional abelian, no non-trivial brackets).

The group action. The Lie algebra $\mathfrak{u}(1)$ integrates to $G = U(1) \cong S^1$. The standard $U(1)$ action on $S^3 \subset \mathbb{C}^2$ is

$$e^{i\theta} \cdot (z_1, z_2) = (e^{i\theta} z_1,\, e^{i\theta} z_2).$$

This action is *free*: no element $e^{i\theta} \neq 1$ has a fixed point on S^3. Therefore H2 (free proper action) is satisfied. H3 is *not* satisfied: $\pi_3(S^3) = \mathbb{Z} \neq 0$.

The Serre long exact sequence. Since the action is free, $S^1 \to S^3 \to Q = S^3/U(1)$ is a principal S^1-bundle. The Serre long exact sequence gives

$$\cdots \to \pi_k(S^1) \to \pi_k(S^3) \to \pi_k(Q) \to \pi_{k-1}(S^1) \to \cdots$$

We evaluate the relevant segment using standard homotopy groups: $\pi_1(S^1) = \mathbb{Z}$, $\pi_k(S^1) = 0$ for $k \geq 2$; $\pi_1(S^3) = 0$, $\pi_2(S^3) = 0$, $\pi_3(S^3) = \mathbb{Z}$.

At $k = 2$:

$$\underbrace{\pi_2(S^1)}_{0} \to \underbrace{\pi_2(S^3)}_{0} \to \pi_2(Q) \to \underbrace{\pi_1(S^1)}_{\mathbb{Z}} \to \underbrace{\pi_1(S^3)}_{0}.$$

By exactness, the sequence $0 \to 0 \to \pi_2(Q) \to \mathbb{Z} \to 0$ forces

$$\pi_2(Q) \;\cong\; \mathbb{Z}.$$

At $k = 1$: the sequence $\cdots \to \pi_1(S^1) \to \pi_1(S^3)$ becomes $\mathbb{Z} \to 0$, and the segment $\pi_1(S^3) \to \pi_1(Q) \to \pi_0(S^1)$ becomes $0 \to \pi_1(Q) \to 0$, giving $\pi_1(Q) = 0$.

Identification. The space Q satisfies $\pi_1(Q) = 0$ (simply connected), $\pi_2(Q) \cong \mathbb{Z}$, and can be identified as homotopy-equivalent to S^2. This is the classical Hopf fibration $S^1 \hookrightarrow S^3 \to S^2$. Without naming $\sim$ or declaring the target geometry, the single generator A and the Serre sequence have forced $Q \simeq S^2$.

Architecture lower bound. Applying the embedding floor from Paper 3: S^2 is a compact 2-manifold, so $\text{dm}(S^2) = 2$. The Whitney embedding theorem gives $\text{emb}(S^2) \leq 4$ and the standard smooth embedding gives $\text{emb}(S^2) = 3$ (the round sphere in $\mathbb{R}^3$). The Menger–Nöbeling lower bound gives $\text{emb}(S^2) \geq 2$; the characteristic class argument (non-trivial π_2, orientable compact surface) gives $\text{emb}(S^2) \geq 3$.

$$\text{emb}(Q) \;\geq\; 3. \tag{2}$$

Consequence. Any encoder $E\colon S^3 \to \mathbb{R}^m$ with $m \leq 2$ is in the embedding floor: it cannot represent the quotient topology faithfully, regardless of the training objective or dataset size. This is derived entirely from the existence of one generator, without declaring $\sim$ or identifying Q geometrically in advance.

Remark 3.7 (Sharpness). The bound $\text{emb}(Q) \geq 3$ is tight: S^2 embeds smoothly in $\mathbb{R}^3$. The Hopf fibration therefore demonstrates that the generator-to-architecture chain can be sharp, not merely a loose inequality.

Remark 3.8 (Covering ambiguity in this example). The Hopf example is free of covering ambiguity. The generator specifies $\mathfrak{u}(1)$, which integrates uniquely to $U(1) \cong S^1$ (no non-trivial covers of S^1 preserve the compact Lie group structure). Corollary 2.10 therefore does not affect this example: the single-generator partial specification is sufficient to run the full chain without ambiguity.

3.4 Structural Example: Molecular Property Prediction

The Hopf fibration example demonstrates the forward chain for a physics-motivated generator. This subsection demonstrates the same chain for a *structural* generator specification — one derived from the measurement geometry of the data-generating process rather than from a conservation law — and shows that the pre-computed hypergraph directly predicts the architectural design choices of a known sequence of empirically validated models.

Generator specification. Molecular property prediction operates on point clouds of atoms in $\mathbb{R}^3$. The structural fact is that a molecule's identity is invariant to rigid motions: rotations, translations, and reflections. This gives a generator specification with six generators: $S = \{A_1, A_2, A_3, T_1, T_2, T_3\}$, where A_i are the $\mathfrak{so}(3)$ rotation generators and T_j are the $\mathbb{R}^3$ translation generators. The known relations are the Euclidean algebra $R_{\text{known}} = \{[A_i, A_j] = \varepsilon_{ijk}A_k,\ [A_i, T_j] = \varepsilon_{ijk}T_k,\ [T_i, T_j] = 0\}$, with type tag $\tau =$ Lie and observation space $\mathcal{E} = (\mathbb{R}^3)^N$ (the N-atom point cloud).

This specification encodes structural knowledge from the measurement geometry, not from Noether's theorem. The generators are known before any molecular data is seen.

Pre-computed hypergraph $\mathcal{H}(G_{\text{spec}})$.

Tier	Algebra $\mathfrak{g}'$	Group G'	Quotient Q'	Floor m^*
0	$\{0\}$	$\{e\}$	$(\mathbb{R}^3)^N$	$3N$
1	$\text{span}(T_j)$	$\mathbb{R}^3$ (translations)	$(\mathbb{R}^3)^N/\mathbb{R}^3$	$3N-3$
2	$\mathfrak{so}(3)$	SO(3)	$(\mathbb{R}^3)^N/\text{SO}(3)$	$3N-3$
3	$\mathfrak{se}(3)$	SE(3)	$(\mathbb{R}^3)^N/\text{SE}(3)$	$3N-6$

The floor at Tier 2 follows from the SO(3) action: the Serre sequence for $\text{SO}(3) \hookrightarrow S^3 \to S^2$ (the Hopf fibration, proved in Section 3.3) gives $\pi_2(Q') \cong \mathbb{Z}$, so the quotient contains a S^2 factor requiring at least 3 dimensions. The floor at Tier 3 reflects the additional 3 degrees of freedom removed by translations.

The closure test confirms the algebra is already closed: all brackets lie in $\text{span}(S)$ by R_{known}. No extension is needed.

What the hypergraph predicts. The hypergraph is pre-computed from the structural specification. Before seeing a single molecule, it certifies:

1. Any architecture operating only on pairwise distances (Tier 0 to Tier 1) cannot be quotient-faithful for SO(3), because pairwise distances are SO(3)-invariant but do not separate all SO(3)-inequivalent configurations (enantiomers, chiral molecules).
2. Tier 2 requires angular features (representations of S^2) in the latent space. The floor $m^* = 3N-3$ is the minimum latent dimension for a quotient-faithful encoder at this level.

3. The covering discriminant at Tier 2 is the SO(3) vs O(3) choice: whether $\exp(2\pi A_i) = I$ (bosonic, SO(3)) or the architecture is sensitive to reflections (O(3), handling chirality).

Correspondence to known architectures. The empirically validated progression in molecular property prediction traces a path up the hypergraph:

SchNet [Schütt et al., 2017] uses only pairwise distances as features. This corresponds to Tier 1 of the hypergraph: translation is removed but SO(3) is not encoded equivariantly. The framework predicts this is insufficient for chirality-sensitive properties.

DimeNet [Gasteiger et al., 2020] adds angular features (bond angles) to the message passing. This corresponds to ascending to Tier 2: SO(3) equivariance is introduced at the message level. The framework predicts this is the minimum architecture for quotient faithfulness at the rotation level.

MACE [Batatia et al., 2022] uses full irreducible representations of SO(3) (and optionally O(3), handling chirality). This corresponds to Tier 2–3 of the hypergraph with the covering discriminant resolved. MACE is the current state-of-the-art molecular force field; the hypergraph predicts it is the minimum architecture that is quotient-faithful for the full SE(3) specification.

Remark 3.9 (Scope of the prediction). The hypergraph predicts the *necessary structural features* of quotient-faithful architectures — angular features at Tier 2, full irreps at Tier 3 — and the minimum latent dimension at each tier. It does not predict every design choice in SchNet, DimeNet, or MACE (message passing schedule, readout function, hyperparameters). The claim is precisely that architectures violating the floor at their tier cannot be quotient-faithful, and that the empirical progression confirms this prediction.

4 The Regular Orbit Quotient: Proofs

This section assembles the proofs of Theorem 2.9, Corollary 2.10, and Proposition 2.11. The arguments draw on classical results in algebraic topology and fibre bundle theory; what is new is their assembly into a chain from generator specification to architecture lower bound. References for the classical components are given at each step.

4.1 Proof of Theorem 2.9

We establish each part in order.

Part (i): The orbit projection is a principal G-bundle.

By hypothesis H2, G acts freely and properly on $\mathcal{X}$. A free proper action of a Lie group on a manifold always yields a principal G-bundle [Lee, 2013, Theorem 21.10]: the orbit projection $q\colon \mathcal{X} \to \mathcal{X}/G = Q$ is a locally trivial fibre bundle with structure group G and typical fibre G. Local triviality follows from the slice theorem for proper free actions [Duistermaat and Kolk, 2012, Theorem 2.3.3].

Remark 4.1 (Why freeness is essential). If the action has fixed points (H2 violated), the orbit space is not locally Euclidean at fixed points, and the projection is not a bundle map. The space $\mathcal{X}/G$ becomes a stratified space: the principal stratum (free orbits) retains bundle structure, while singular strata (fixed-point orbits) do not. See Section 5 for the stratified extension.

Part (ii): The Serre long exact sequence and its collapse.

The principal G-bundle $G \hookrightarrow \mathcal{X} \xrightarrow{q} Q$ is in particular a Serre fibration with fibre G, base Q, and total space $\mathcal{X}$. The Serre long exact sequence of homotopy groups reads [Hatcher, 2002, Theorem 4.41]:

$$\cdots \to \pi_k(G) \xrightarrow{i_*} \pi_k(\mathcal{X}) \xrightarrow{q_*} \pi_k(Q) \xrightarrow{\partial} \pi_{k-1}(G) \to \cdots \to \pi_0(\mathcal{X}) \to 0. \tag{3}$$

Under hypothesis H3, $\mathcal{X}$ is contractible, so $\pi_k(\mathcal{X}) = 0$ for all $k \geq 1$. The sequence (3) therefore reduces, at each $k \geq 1$, to the short exact segment

$$0 = \pi_k(\mathcal{X}) \to \pi_k(Q) \xrightarrow{\partial} \pi_{k-1}(G) \to \pi_{k-1}(\mathcal{X}) = 0, \tag{4}$$

from which exactness gives the isomorphism

$$\pi_k(Q) \;\cong\; \pi_{k-1}(G) \qquad \text{for all } k \geq 1. \tag{5}$$

Remark 4.2 (Contractibility of $\mathbb{R}^n$). The hypothesis H3 is satisfied for $\mathcal{X} = \mathbb{R}^n$, which is contractible via the straight-line homotopy $H(x,t) = (1-t)x$. It is also satisfied for any open convex subset of $\mathbb{R}^n$. Most machine learning observation spaces (sensor readings, image pixel arrays, state vectors) are modelled as Euclidean, so H3 is the natural default assumption. When $\mathcal{X}$ is not contractible (e.g. $\mathcal{X} = S^3$ as in the Hopf example), the full sequence (3) must be used; see Remark 3.6.

Part (iii): Homotopy groups constrained by the generator specification.

The isomorphism (5) gives $\pi_k(Q) \cong \pi_{k-1}(G)$. The homotopy groups $\pi_*(G)$ depend on the global topology of G, which is constrained by the generator specification G_{spec} as follows.

The Lie algebra $\mathfrak{g} = \overline{S}$ (the Lie algebraic closure of the generator symbols S) determines a unique simply-connected Lie group $\tilde{G}$, called the *universal cover* of G, together with a discrete central subgroup $\Gamma \leq Z(\tilde{G})$ such that $G \cong \tilde{G}/\Gamma$ [Hall, 2015, Theorem 5.52]. The choice of Γ is the covering ambiguity of Corollary 2.10.

For $k \geq 2$: the covering map $\tilde{G} \to G$ induces isomorphisms $\pi_k(\tilde{G}) \cong \pi_k(G)$, so $\pi_k(G)$ for $k \geq 2$ is fully determined by $\mathfrak{g}$ alone.

For $k = 1$: $\pi_1(G) \cong \Gamma$ (since $\tilde{G}$ is simply connected), which is not determined by $\mathfrak{g}$ alone. It depends on the global identification of G. A partial specification with $R_{\text{known}} = \varnothing$ therefore leaves $\pi_1(G)$ undetermined within the finite set of admissible Γ.

Combining with (5): $\pi_k(Q)$ is fully determined for $k \geq 3$, and is determined up to covering ambiguity for $k = 2$ (since $\pi_2(Q) \cong \pi_1(G) \cong \Gamma$).

Part (iv): Architecture lower bound.

Applying the embedding floor function from Paper 3 [Farkas, 2026c]: for any compact metric space Q with covering dimension $\mathrm{dm}(Q) = n$, the Menger–Nöbeling theorem gives $\mathrm{emb}(Q) \leq 2n + 1$. For smooth manifolds the Whitney embedding theorem gives $\mathrm{emb}(Q) \leq 2n$. For specific quotient topologies, sharper lower bounds are available from the homotopy data via characteristic class obstructions and the Haefliger improvement; these are tabulated in Paper 3 for all compact 3-manifolds and for the principal families appearing in Lie group quotients.

Setting $F(\pi_1(G), \pi_2(G), \mathrm{dm}(Q))$ to the appropriate entry in this table (or to the Menger–Nöbeling lower bound when the family is not catalogued), Part (iv) follows. $\square$

4.2 Proof of Corollary 2.10

The Lie algebra $\mathfrak{g}$ determines the universal cover $\tilde{G}$ uniquely up to isomorphism [Hall, 2015, Corollary 5.7]. The possible groups with Lie algebra $\mathfrak{g}$ are the quotients $\tilde{G}/\Gamma$ for subgroups $\Gamma \leq Z(\tilde{G})$. The centre $Z(\tilde{G})$ is finite for compact semisimple $\tilde{G}$ (e.g. $Z(\mathrm{SU}(2)) = \mathbb{Z}/2$), so the set of admissible groups is finite.

For $k \geq 2$: the long exact sequence of the covering $\tilde{G} \to G$ gives $\pi_k(\tilde{G}) \cong \pi_k(G)$ (since $\pi_k(\text{point}) = 0$ for $k \geq 1$ and discrete fibres have no higher homotopy), so $\pi_k(G)$ is determined by $\mathfrak{g}$ for $k \geq 2$.

For $k = 1$: $\pi_1(G) \cong \Gamma$ varies over admissible central subgroups. For the canonical example, $\mathfrak{g} = \mathfrak{so}(3) \cong \mathfrak{su}(2)$, $\tilde{G} = \mathrm{SU}(2)$, and the admissible groups are $\mathrm{SU}(2)$ (with $\Gamma = \{e\}$, $\pi_1 = 0$) and $\mathrm{SO}(3)$ (with $\Gamma = \mathbb{Z}/2$, $\pi_1 = \mathbb{Z}/2$).

Via the isomorphism $\pi_2(Q) \cong \pi_1(G)$ from Theorem 2.9(ii), the covering ambiguity in $\pi_1(G)$ induces an ambiguity in $\pi_2(Q)$:

$$\pi_2(Q) \cong \begin{cases} 0 & \text{if } G = \mathrm{SU}(2), \\ \mathbb{Z}/2 & \text{if } G = \mathrm{SO}(3). \end{cases} \tag{6}$$

The embedding floors in the two cases may differ: a quotient with $\pi_2(Q) = \mathbb{Z}/2$ (torsion) may require a higher ambient dimension than one with $\pi_2(Q) = 0$.

The sharpest lower bound uses the simply connected cover $\tilde{G}$ (giving $\pi_1(G) = 0$, hence $\pi_2(Q) = 0$ from this source); the weakest uses the quotient with the largest $|\Gamma|$. Both are valid lower bounds on $\mathrm{emb}(Q)$; the gap between them is quantified by the characteristic class obstruction from the torsion in $\pi_1(G)$.

The covering ambiguity is resolved when R_{known} contains global relations, such as $\exp(2\pi A_i) = e$ (which forces $G = \mathrm{SO}(3)$ over $\mathrm{SU}(2)$ for the angular momentum generators), or equivalently when physical domain knowledge (e.g. integer vs. half-integer spin) selects the admissible Γ. □

4.3 Proof of Proposition 2.11

Let $Q \in \mathcal{Q}(G_{\text{spec}}, \mathcal{X})$ be any admissible quotient with $\mathrm{emb}(Q) > m$. Suppose for contradiction that a continuous quotient-faithful encoder $E\colon \mathcal{X} \to \mathbb{R}^m$ exists. Then the induced map $\tilde{E}\colon Q \to \mathbb{R}^m$ is a topological embedding of Q into $\mathbb{R}^m$. But $\mathrm{emb}(Q) > m$ means no such embedding exists. Contradiction.

Since this holds for every $Q \in \mathcal{Q}(G_{\text{spec}}, \mathcal{X})$, no encoder into $\mathbb{R}^m$ is quotient-faithful for any admissible quotient. □

Remark 4.3 (The embedding floor in the generator-constrained setting)*.* Proposition 2.11 is the generator-constrained version of the embedding floor proved in Paper I [Farkas, 2026a]. Paper I establishes the trap for a declared quotient Q; here the trap applies to the entire admissible class without declaring Q. The practical consequence is the same: the latent dimension m must satisfy $m \geq \min_{Q \in \mathcal{Q}} \mathrm{emb}(Q)$ or the architecture is guaranteed to fail.

4.4 The Two Physical Regimes

In physical applications the generator specification G_{spec} is not produced by guessing over a large space of possible groups. Noether's theorem provides a direct map from conservation laws to the generator catalog:

$$\text{Conservation law} \xrightarrow{\text{Noether}} \text{Conserved charge} \xrightarrow{\exp} \text{Lie group family.} \tag{7}$$

The catalog of Lie groups that can appear as physical symmetry groups is small and well-understood. The covering ambiguity is typically resolved by physical observables: a system admitting half-integer representations (fermions) has symmetry group $\mathrm{SU}(2)$; a system with only integer representations (bosons) has $\mathrm{SO}(3)$. This is not inference from data but a consequence of the representation theory of the rotation group.

We therefore distinguish two regimes for the GCD training scheme of Section 9:

Regime A (physics known). The dynamics of the system are understood and Noether's theorem has been applied. The generator catalog is derived from conservation laws; the prior over admissible groups is informative and finite. Bayesian model selection over this catalog is efficient, principled, and does not require topological data analysis. The covering ambiguity is resolved by physical domain knowledge. The principled derivation of generators from energy structure is formalised by Capucci et al. [2025]: their reaction structure $R\colon T^*X \to TX$ (Definition 3) is the cotangent-to-tangent map that encodes the physical law, and the energy functional $E\colon X \to \mathbb{R}$ (Definition 5) is the source from which the generator catalog is derived. Hamiltonian mechanics and gradient descent differ only in their reaction structure [Capucci et al., 2025, Introduction]. This is the appropriate regime for physical simulation, molecular dynamics, robotics, and any other application where the governing equations are known. The sample efficiency benefit of encoding a physics-derived generator catalog is empirically demonstrated by Gong et al. [2022]: LorentzNet, which encodes the known $\mathrm{SO}(1,3)^+$ Lorentz symmetry from physics into the architecture, substantially outperforms a learned architecture at 0.5% of training data, with the gap widening as data is reduced.

Regime B (physics unknown). No conservation laws are available. The data-generating process may have hidden symmetries that have not been characterised theoretically. Here, the generator catalog cannot be derived analytically and must be constructed from data. Persistent homology pre-screening of the initial latent provides topology evidence before any group structure is assumed; algebraic closure testing then checks whether the learned generators are consistent with that evidence; and a topological autoencoder loss [Moor et al., 2020] maintains the measured topology during training.

The theoretical backbone — Definition 2.1, Theorem 2.9, Proposition 2.11 — is identical in both regimes. The generator specification G_{spec} accommodates both: in Regime A, R_{known} is populated by Noether analysis and the covering ambiguity is resolved by physical reasoning; in Regime B, R_{known} starts empty and the evidence set $\mathcal{E}$ carries topological data.

5 Boundary Cases: When the Regularity Hypotheses Fail

Theorem 2.9 and its corollaries rest on three hypotheses (H1–H3). This section examines what happens when each is relaxed, and states what can still be salvaged.

5.1 Non-free Actions (H2 Violated)

When the action of G on $\mathcal{X}$ has fixed points, the orbit projection $\mathcal{X} \to \mathcal{X}/G$ is no longer a bundle map and the Serre sequence does not apply globally. The orbit space acquires a stratified structure:

Definition 5.1 (Orbit-type stratification)**.** Let G act continuously on $\mathcal{X}$. For each subgroup $H \leq G$, the H*-stratum* is

$$\mathcal{X}_H \;=\; \{x \in \mathcal{X} \mid G_x = H\},$$

where $G_x = \{g \in G \mid g \cdot x = x\}$ is the isotropy group of x. The *principal stratum* $\mathcal{X}_{\{e\}}$ consists of points with trivial isotropy; it is the free-orbit locus and is dense in $\mathcal{X}$ when G is compact.

Proposition 5.2 (Bounds on the principal stratum)**.** *Let G be a connected compact Lie group acting on $\mathcal{X}$ (H3 retained). On the principal stratum $\mathcal{X}_{\{e\}}$, the restriction of the orbit projection is a principal G-bundle. Therefore Theorem 2.9 applies to the restricted quotient $\mathcal{X}_{\{e\}}/G$, and the embedding floor*

$$\mathrm{emb}\big(\mathcal{X}_{\{e\}}/G\big) \;\geq\; F(\pi_*(G), \mathrm{dm})$$

holds. This provides a lower bound on $\mathrm{emb}(\mathcal{X}/G)$*, since any encoder into* $\mathbb{R}^m$ *that is faithful on the full quotient must also be faithful on the principal-stratum quotient.*

Remark 5.3 (The singular strata). Points in singular strata $\mathcal{X}_H$ for $H \neq \{e\}$ have orbits of smaller dimension $\dim(G) - \dim(H)$ and the local quotient structure near these points requires the language of orbifolds or, more generally, differentiable stacks. The topology of the full orbit space $\mathcal{X}/G$ at singular points can be more complex than the principal-stratum analysis suggests. For the purposes of architecture lower bounds, Proposition 5.2 gives a conservative but rigorous floor.

The most common physical case is $G = \mathrm{SO}(3)$ acting on $\mathcal{X} = \mathbb{R}^3$: the origin is a fixed point and the principal stratum is $\mathbb{R}^3 \setminus \{0\}$. The restricted quotient is $(\mathbb{R}^3 \setminus \{0\})/\mathrm{SO}(3) \simeq S^2$, giving emb ≥ 3 on the free-orbit locus.

5.2 Semantic Coarsenings (Beyond Orbit Equivalence)

In representation learning, the lawful quotient $Q = \mathcal{X}/{\sim}$ is determined by the *task-relevant* equivalence relation, which need not coincide with the orbit equivalence $\sim_G$ induced by the symmetry group. Two cases arise:

Case 1: $\sim$ refines $\sim_G$. The admissible relation identifies fewer points than orbit equivalence. The quotient $Q = \mathcal{X}/{\sim}$ is a further quotient of the principal bundle, and the bundle structure no longer applies directly. The tower $\mathcal{X} \to \mathcal{X}/G \to \mathcal{X}/{\sim}$ is a sequence of quotient maps, and the topology of Q depends on the second map as well as the first. In this case Theorem 2.9 gives a lower bound on the intermediate quotient $\mathcal{X}/G$; additional information about the coarsening map is needed to bound $\mathrm{emb}(Q)$ directly.

Case 2: $\sim$ agrees with $\sim_G$. This is the orbit-quotient case, where Theorem 2.9 applies without qualification. For Regime A (physics known), Noether's theorem typically identifies the orbit equivalence as the natural admissible relation, and this case is the most common in practice.

Remark 5.4. The gap between Case 1 and Case 2 is the source of one of the primary open problems in the framework: when the task map $t\colon Q \to Y$ is non-injective, the graded compression pushes toward a further quotient of Q (the label representation $E_{\text{label}} = t \circ q$), which is a coarsening of the orbit quotient. Theorem 2.9 bounds the orbit quotient; controlling the graded compression requires the additional machinery of Section 9.

5.3 Non-contractible Observation Spaces (H3 Violated)

When $\mathcal{X}$ is not contractible, the Serre long exact sequence does not simplify to the isomorphisms of Theorem 2.9(ii). The full sequence

$$\cdots \to \pi_k(G) \to \pi_k(\mathcal{X}) \to \pi_k(Q) \to \pi_{k-1}(G) \to \cdots$$

must be used, with $\pi_*(\mathcal{X})$ as an additional input.

This is not a failure of the framework; it is a regime in which more information is required. The Hopf fibration example of Section 3.3 demonstrates that the full sequence remains tractable and sharp even without contractibility: $\mathcal{X} = S^3$ has $\pi_3(S^3) = \mathbb{Z}$, and the full sequence correctly identifies $\pi_2(Q) \cong \mathbb{Z}$ (forcing $Q \simeq S^2$) while the higher groups are determined by the non-trivial $\pi_3(S^3)$.

Remark 5.5 (Extension to non-contractible $\mathcal{X}$). The generator specification G_{spec} in Definition 2.1 already accommodates this case: the evidence set $\mathcal{E}$ can include topological data about $\mathcal{X}$ (e.g. from persistent homology pre-screening) that supplies the needed $\pi_*(\mathcal{X})$ information. In Regime A, the topology of $\mathcal{X}$ is typically known from physics (e.g. $\mathcal{X} = S^3$ for unit quaternions); in Regime B it must be estimated from data.

6 Discrete Groups and Covering-Type Quotients

When the generator type is $\tau =$ discrete, the generating group $G = \langle S \mid R \rangle$ is a discrete group given by a finite presentation. The topological machinery of Section 3 adapts as follows.

Proposition 6.1 (Discrete covering quotient constraint)**.** *Let G be a discrete group acting freely and properly discontinuously on a simply connected topological space $\mathcal{X}$. Then:*

(i) *The orbit projection* $q\colon \mathcal{X} \to Q = \mathcal{X}/G$ *is a regular covering map with deck group* G.

(ii) *The long exact sequence of the covering gives* $\pi_1(Q) \cong G$ *and* $\pi_k(Q) \cong \pi_k(\mathcal{X})$ *for* $k \geq 2$.

(iii) *The group presentation* $\langle S \mid R \rangle$ *therefore constrains* $\pi_1(Q)$*, and via the embedding floors of Paper 3, constrains* $\operatorname{emb}(Q)$.

Proof. Part (i) is the standard covering space theorem for free properly discontinuous actions on path-connected, locally path-connected, semi-locally simply connected spaces [Hatcher, 2002, Theorem 1.38]. Since $\mathcal{X}$ is simply connected, it is the universal cover of Q, and the deck transformation group is $\pi_1(Q) \cong G$. Parts (ii) and (iii) follow from the long exact sequence of the fibration $G \hookrightarrow \mathcal{X} \to Q$ with G discrete (so $\pi_k(G) = 0$ for $k \geq 1$), which gives $\pi_k(Q) \cong \pi_k(\mathcal{X})$ for $k \geq 2$ and the covering isomorphism $\pi_1(Q) \cong G$ for $k = 1$. □ □

Remark 6.2 (Scope and limitations). Proposition 6.1(iii) gives a π_1-derived bound. The step from $\pi_1(Q)$ to $\operatorname{emb}(Q)$ is family-dependent: not every group G forces a high embedding dimension. For example, $G = \mathbb{Z}$ gives $Q \simeq S^1$ with $\operatorname{emb}(Q) = 2$; $G = \mathbb{Z}/n$ gives a lens space with $\operatorname{emb}(Q) \geq 3$ for $n \geq 2$. The bound is non-trivial whenever $\pi_1(Q) \neq 0$ imposes topological constraints that prevent embedding in $\mathbb{R}^1$ (traversal solutions), which is the practically relevant case.

Note also that the claim $\pi_k(Q) \cong \pi_k(\mathcal{X})$ for $k \geq 2$ requires $\mathcal{X}$ to be simply connected. For non-simply-connected $\mathcal{X}$, the full long exact sequence must be used, as in Section 5.3.
Remark 6.3 (Polynomial comonad interpretation). By a result of Ahman and Uustalu [Spivak, 2025], a small category C is identified up to isomorphism with a polynomial comonad $k_{\mathsf{C}} = \sum_{c\colon \mathrm{Ob}\,\mathsf{C}} y^{\mathsf{C}[c]}$ on **Set**, where $\mathsf{C}[c]$ is the set of outgoing morphisms from c. A discrete group G, viewed as a one-object category $\mathbf{B}G$, corresponds to the polynomial $y^{|G|}$ with comultiplication encoding group multiplication.

Under this identification, a generator specification $G_{\text{spec}} = (S, R_{\text{known}}, \text{discrete}, \mathcal{E})$ partially specifies the polynomial comonad $k_{\mathbf{B}G}$: the set S determines the output arity $|G|$, and the known relations R_{known} constrain the comultiplication. The admissible quotient class $\mathcal{Q}(G_{\text{spec}}, \mathcal{X})$ of Definition 2.3 is therefore the class of all instances consistent with this partially specified polynomial comonad. This is a categorical formulation of quotient constraint without naming $\sim$ explicitly.

7 Formal Axioms as Abstract Generators

The generator-constrained framework extends beyond physical symmetry groups to settings where the generators are formal axioms and the relations are inference rules. This section makes that extension precise using two complementary frameworks: Spivak's categorical lifting problems [Spivak, 2014] and the Barkeshli–Douglas–Freedman (BDF) proof hypergraph [Barkeshli et al., 2026].

7.1 The Lifting Problem Formulation

A *quotient type* in formal mathematics requires that a function $f'\colon A \to B$ descends to a well-defined function $\bar{f}\colon A/R \to B$ if and only if it respects the equivalence relation R. The descent condition is:

$$\forall x, y.\ R(x, y) \ \Rightarrow\ f'(x) = f'(y).$$

In Spivak's framework [Spivak, 2014], this is precisely a *lifting constraint*. Modelling the data space A as a database instance $\pi\colon I \to S$ and the equivalence relation R as a constraint pair (m, n) in the sense of Spivak [2014, Definition 3.1.1], the descent condition states that π satisfies the lifting constraint (m, n): for every solid-arrow commutative diagram

$$\begin{tikzcd} W \arrow[r, "p"] \arrow[d, "m"'] & I \arrow[d, "\pi"] \\ R \arrow[r, "n"'] \arrow[ur, dashed, "\ell"] & S \end{tikzcd}$$

a lift ℓ exists. The admissibility declaration $\sim$ of Paper 2 is exactly this lifting constraint: a function may descend to the quotient if and only if it satisfies the constraint.

This is not merely an analogy. The Spivak framework and our Definition 2.3 share the same mathematical structure:

- A generator specification $G_{\text{spec}} = (S, R_{\text{known}}, \tau, \mathcal{E})$ corresponds to a database schema in the sense of Schultz et al. [2016, Definition 5.6]: the generator set S is the graph G of the category presentation, the known relations R_{known} are the path equations E, and the type assignment τ is the algebraic profunctor signature Υ.
- The admissible quotient class $\mathcal{Q}(G_{\text{spec}}, \mathcal{X})$ corresponds to the set of schema instances satisfying those constraints.
- The covering ambiguity of Corollary 2.10 corresponds to the non-uniqueness of lifts when the constraint set is underdetermined.

Furthermore, Spivak [2014, Example 2.3.7] shows that a covering of groupoids is precisely a surjective discrete opfibration. The SO(3)/SU(2) ambiguity of Corollary 2.10 is the topological form of the non-uniqueness of surjective discrete opfibrations over the same base, which is in turn the non-uniqueness of lifts in an underdetermined constraint system.

7.2 The BDF Connection

Barkeshli et al. [2026] formalise a mathematical discovery system using a universal proof hypergraph $\mathcal{U}$. Axioms are the root nodes (generators) and inference rules are the hyperedges (relations). In their framework, defining a function on a quotient type A/R requires discharging a proof obligation they call the *Consistency Hyperedge* (§2.5 of their paper):

$$\forall x, y.\ R(x, y) \ \Rightarrow\ f'(x) = f'(y).$$

This is, by the analysis of Section 7.1, a lifting constraint in the sense of Spivak [2014]. The connection is therefore not an analogy but a mathematical identity: the BDF Consistency Hyperedge is the lifting-constraint form of the admissibility declaration.

What this establishes. A neural theorem prover learning representations of mathematical objects faces the same three failure modes as a physical representation learner:

- *Traversal trap:* the network may memorise specific proof paths (finite sample) without learning the lawful quotient structure of the type.
- *Compression trap:* if the task is only to check provability, task loss drives the representation toward $E_{\text{label}} = t \circ q$, collapsing semantically distinct types whenever they have the same provability status.
- *Dimension trap:* the type A/R may require a higher embedding dimension than

the network's latent space, in which case quotient-faithful encoding is impossible regardless of training.

The Consistency Hyperedge is the admissibility source that prevents these traps: it enforces the descent condition that makes f' factor through A/R, which is the categorical form of a Route A declaration.

What this does not establish. The topological chain of Sections 3–6 does not automatically transfer to formal languages. The Serre long exact sequence requires a topological space and a continuous group action; a proof hypergraph is a combinatorial object, and its "topology" (if any) must be constructed separately. Whether the classifying space $B\langle S \mid R\rangle$ of the group presented by axioms and inference rules provides useful topological bounds on the embedding dimension of the type quotient remains an open problem; see Open Problem 10.

The correct epistemic status. The BDF connection establishes that:

1. the admissibility framework of the present series applies to formal systems, not only physical ones;
2. the Consistency Hyperedge is the exact categorical form of the admissibility declaration;
3. the lifting-constraint framework of Spivak [2014] provides the common language in which both the physical and formal cases can be stated.

The topological bounds (dimension floors, Serre sequence) apply in the physical and discrete cases where the group action on a topological space is available. Extending them to formal type quotients requires additional structure and constitutes a research programme, not a proved result.

8 The Differentiable Generator Bottleneck

The theoretical chain of Sections 2–7 is kinematic: it constrains which quotient class is admissible given a generator specification, and thereby constrains which architecture families are candidates. This section describes the differentiable implementation of the generator constraint — the horizontal step of Section 3.1 turned into a trainable module.

8.1 The Generator Matrix Parametrisation

Let the data space $\mathcal{X}$ carry a linear action of G and let the encoder be $E_\theta \colon \mathcal{X} \to \mathbb{R}^m$. We parametrise the generators of G as a learnable set of $d \times d$ matrices $\mathcal{M} = \{M_1, \ldots, M_k\} \subset \mathfrak{gl}(d)$, where d is the dimension of the data representation. Each M_i is intended to approximate the Lie algebra generator corresponding to the i-th physical or structural symmetry.

The group action on the latent space is approximated by the matrix exponential:

$$T_\alpha(z) = \exp\Big(\sum_{i=1}^{k} \alpha_i M_i\Big) z, \qquad z \in \mathbb{R}^m,\ \alpha \in \mathbb{R}^k.$$

This is differentiable in both M_i and α_i, making the generator matrices trainable by gradient descent.

Definition 8.1 (Generator bottleneck)**.** Given an encoder E_θ and a generator set $\mathcal{M}$, the *generator bottleneck* is the equivariance constraint

$$E_\theta(T_\alpha(x)) = \exp\Big(\sum_i \alpha_i M_i\Big) E_\theta(x)$$

required to hold for all $x \in \mathcal{X}$ and all α. A representation satisfying this constraint is equivariant with respect to the parametrised group action.

The bottleneck enforces that the encoder commutes with the group action up to the matrix exponential. Crucially, this is a constraint on the representation rather than on the architecture: any encoder family can in principle satisfy it, but only if the latent dimension m is large enough to accommodate the quotient topology (Proposition 2.11).

8.2 The Equivariance Loss

The generator bottleneck is enforced via a differentiable loss term.

Definition 8.2 (Equivariance loss)**.** Let $\mathcal{D} = \{x_j\}_{j=1}^n \subset \mathcal{X}$ be a finite sample and let $\mathcal{A}$ be a distribution over group parameters. The *equivariance loss* is

$$\mathcal{L}_{\text{equiv}}(\theta, \mathcal{M}) \;=\; \mathbb{E}_{x\sim\mathcal{D},\, \alpha\sim\mathcal{A}} \Big\| E_\theta\big(T_\alpha(x)\big) - \exp\Big(\sum_i \alpha_i M_i\Big) E_\theta(x) \Big\|^2 .$$

Minimising $\mathcal{L}_{\text{equiv}}$ jointly over θ and $\mathcal{M}$ trains the encoder to be equivariant with respect to the parametrised group. When $\mathcal{L}_{\text{equiv}} = 0$, the encoder exactly satisfies the generator bottleneck of Definition 8.1.

Remark 8.3 (Degenerate zero solution). The equivariance loss admits a degenerate global minimiser: if $E_\theta(x) = 0$ for all x, then $\mathcal{L}_{\text{equiv}} = 0$ trivially, since $\exp(\sum_i \alpha_i M_i) \cdot 0 = 0$. The closure loss is also zero at this point. This degenerate solution satisfies all structural constraints while destroying all topological content of the representation; it is a concrete realisation of traversal ambiguity in the sense of Paper I.

To prevent radial collapse, a geometric regulariser targeting the natural shell radius of a D-dimensional Gaussian can be added:

$$\mathcal{L}_{\text{radial}} \;=\; -\Big[\Big(\tfrac{1}{2} - \tfrac{1}{D}\Big)\log r^2 - \tfrac{1}{2} r^2\Big], \qquad r^2 = \|E_\theta(x) - c_y\|^2 / D,$$

where c_y is the centroid of class y. The $-\frac{1}{2}r^2$ term is Gaussian gravity pulling the representation inward; the $\log r^2$ term models the expanding volume of the hypersphere and prevents collapse. The equilibrium is $r^2 \approx 1 - 2/D$, the natural shell radius [Pisoni, 2026].[1]

Remark 8.4 (Relation to prior equivariant architectures). Convolutional neural networks [LeCun et al., 1998], group-equivariant CNNs [Cohen and Welling, 2016], and homomorphism-equivariant encoders [Keurti et al., 2023] all enforce equivariance by fixing the group G in the architecture before training. The generator bottleneck parametrises G through $\mathcal{M}$ and learns it from data. This is the difference between a Route A declaration (architecture enforced) and a learnable Route A initialisation (architecture discovered via the bottleneck).

8.3 The Algebraic Closure Loss

The equivariance loss trains $\mathcal{M}$ to explain the observed transformations, but says nothing about whether $\mathcal{M}$ closes under the Lie bracket. Algebraic closure (Section 3.1) is a necessary condition for the matrices to genuinely represent a Lie algebra.

[1] Pisoni [2026] exists at time of writing as a blog post only. The regulariser is an independently motivated construction; the citation is for priority attribution.

Definition 8.5 (Closure loss)**.** Let $P_{\mathcal{M}}$ denote orthogonal projection onto the linear span of $\mathcal{M}$. The *closure loss* is

$$\mathcal{L}_{\text{closure}}(\mathcal{M}) \;=\; \sum_{i<j} \big\| [M_i, M_j] - P_{\mathcal{M}}([M_i, M_j]) \big\|_F^2,$$

where $[M_i, M_j] = M_i M_j - M_j M_i$ is the matrix commutator.

When $\mathcal{L}_{\text{closure}} = 0$, every pairwise bracket lies in $\operatorname{span}(\mathcal{M})$, so $\mathcal{M}$ spans a Lie subalgebra of $\mathfrak{gl}(d)$. By Proposition 3.4, this determines G up to the covering ambiguity of Corollary 2.10.

Remark 8.6 (Sufficiency gap). $\mathcal{L}_{\text{closure}} = 0$ is a *necessary* condition for $\mathcal{M}$ to span a genuine Lie algebra, but it is not sufficient for quotient faithfulness. The map from generators to admissible quotient is constrained by the Serre sequence (Section 3.2), but whether the trained encoder achieves a quotient-faithful representation depends on additional conditions not captured by $\mathcal{L}_{\text{closure}}$ alone; see Open Problem 10.

8.4 The Dimension Floor Check

After training on $\mathcal{L}_{\text{equiv}} + \lambda\,\mathcal{L}_{\text{closure}}$, the learned $\mathcal{M}$ can be matched against the architecture obstruction of Proposition 2.11.

Proposition 8.7 (Architecture obstruction, restatement)**.** *If the learned generator set $\mathcal{M}$ is consistent with a Lie group G acting on $\mathcal{X}$, and if the Serre sequence gives $\operatorname{emb}(Q) \geq m^* > m$ for all $Q \in \mathcal{Q}(G_{\text{spec}}, \mathcal{X})$, then no quotient-faithful encoder into $\mathbb{R}^m$ exists regardless of θ. The architecture must be expanded to latent dimension at least m^* before training can proceed.*

This is not a convergence result. It is a *pre-training certificate*: if the dimension floor exceeds the current latent dimension, training to $\mathcal{L}_{\text{equiv}} \to 0$ cannot produce a quotient-faithful encoder and will instead produce a task-map collapse or traversal ambiguity in the sense of Paper I.

The check requires matching the learned $\mathcal{M}$ against the topology library of known $(\mathfrak{g}, \mathcal{X})$ pairs for which $\operatorname{emb}(Q)$ has been computed. This matching is family-dependent. For the cases covered by Paper 3 (Hopf fibrations, lens spaces, flag manifolds), the bounds are available analytically. For cases outside the library, the topology must be estimated via persistent homology [Moor et al., 2020] applied to $E_\theta(\mathcal{X})$ as an intermediate diagnostic.

9 The Generator-Constrained Discovery Scheme

The theoretical chain of Sections 2–7 is kinematic. This section translates it into a complete operational procedure. The central insight is a clean separation between two computations that have different characters: the *algebraic pre-computation* of the search space from the generator specification alone, and the *statistical localisation* of the data on that pre-computed space. These alternate in an outer loop that extends the search space when the data cannot be explained by the current map.

9.1 The Generator Hypergraph

Given a generator specification $G_{\text{spec}} = (S, R_{\text{known}}, \tau, \mathcal{E})$, define the *generator hypergraph* $\mathcal{H}(G_{\text{spec}})$ as follows, entirely from algebraic and topological computation, before any data is seen.

Nodes. Each node of $\mathcal{H}$ is a subalgebra $\mathfrak{g}' \leq \mathfrak{g} = \mathrm{span}(S)$, together with:

- its *tier* $\ell = \dim \mathfrak{g}'$ (dimension as a vector space),
- its *embedding floor* $m^*(\mathfrak{g}') = \mathrm{emb}(\mathcal{X}/G')$ computed from the Serre sequence of the principal bundle $G' \hookrightarrow \mathcal{X} \to \mathcal{X}/G'$, where G' is the connected subgroup of G with Lie algebra $\mathfrak{g}'$.

Edges. There is a directed edge $\mathfrak{g}'_1 \to \mathfrak{g}'_2$ whenever $\mathfrak{g}'_1 \leq \mathfrak{g}'_2$ and $\dim \mathfrak{g}'_2 = \dim \mathfrak{g}'_1 + 1$ (adjacent tiers in the subgroup lattice). The hyperedges encode the inclusion order; additional hyperedges may be added by the domain prior (see Section 9.6).

Tier structure. The nodes are stratified by tier:

$$\text{Tier } 0 \;=\; \{\{0\}\}, \quad \text{Tier } 1 \;=\; \{\mathrm{span}(A_i) : A_i \in S\}, \quad \ldots, \quad \text{Tier } k \;=\; \{\mathfrak{g}\}.$$

By the dimension monotonicity of the Serre sequence (Appendix B), the floor m^* is non-increasing as tier increases: a larger acting group produces a simpler quotient with a lower or equal embedding floor.

Computational cost. Constructing $\mathcal{H}(G_{\mathrm{spec}})$ requires $O(k^2)$ commutator evaluations to test closure of all generator pairs, and one Serre sequence computation per node to assign floors. Both steps are data-free and polynomial in $k = |S|$. The result is a finite, labelled directed acyclic graph that encodes the complete space of admissible quotients consistent with G_{spec}.

Remark 9.1 (The map is complete within G_{spec}). $\mathcal{H}(G_{\mathrm{spec}})$ contains every subalgebra of $\mathfrak{g}$ and therefore every admissible quotient in $\mathcal{Q}(G_{\mathrm{spec}}, \mathcal{X})$. It does not contain quotients requiring generators outside $\mathfrak{g}$. If the true symmetry group has generators not in $\mathrm{span}(S)$, the correct node is absent from the map; this is the extension case of Section 9.5.

9.2 Localisation on the Pre-computed Map

Once $\mathcal{H}(G_{\mathrm{spec}})$ is constructed, the data $\mathcal{D}$ does exactly one thing: it assigns a posterior probability to each node.

$$p(\mathfrak{g}' \mid \mathcal{D}) \;\propto\; p(\mathcal{D} \mid \mathfrak{g}')\, p(\mathfrak{g}') \tag{8}$$

where $p(\mathcal{D} \mid \mathfrak{g}')$ is the likelihood of the data under the hypothesis that the true symmetry algebra is $\mathfrak{g}'$, approximated by the equivariance loss, and $p(\mathfrak{g}')$ is the prior over nodes, which may be uniform or informed by the domain prior.

The statistical problem is therefore *classification on a pre-labelled finite graph*, not search over an unknown space. The hypothesis class is $\mathcal{H}(G_{\mathrm{spec}})$, fixed before training. By the finite hypothesis class bound, the sample complexity of identifying the correct node is $O(\log |\mathcal{H}_{\mathrm{prior}}|)$, where $|\mathcal{H}_{\mathrm{prior}}|$ is the number of nodes consistent with the domain prior — which is substantially smaller than the full lattice when the prior is informative (fractal structure, energy cascade, sequence prior).

The practical implementation of this posterior update is the four-phase GCD loop of Section 9.4.

9.3 Worked Example: Two Independent Factors

This subsection works through the complete hypergraph pre-computation for a structural generator specification arising in disentangled representation learning, and demonstrates the extension procedure when the initial map is incomplete.

Setup. Let $\mathcal{X} = S^3 \times S^3$ (two independent periodic factors, e.g. two independent angles in a robotic arm, or two independent latent codes in a generative model). The structural prior is that the two factors are *independent*: neither influences the other. This gives the generator specification:

$$S = \{A_1, A_2\}, \quad R_{\text{known}} = \{[A_1, A_2] = 0\}, \quad \tau = \text{Lie},$$

where A_1 is the infinitesimal $U(1)$ rotation of factor 1 and A_2 of factor 2. The known relation $[A_1, A_2] = 0$ encodes independence.

Pre-computed hypergraph $\mathcal{H}(G_{\text{spec}})$.

Tier	Node	Group G'	Quotient Q'	Floor m^*
0	$\mathfrak{g}_0 = \{0\}$	$\{e\}$	$S^3 \times S^3$	6
1	$\mathfrak{g}_1 = \text{span}(A_1)$	$U(1)$	$S^2 \times S^3$	5
1	$\mathfrak{g}_2 = \text{span}(A_2)$	$U(1)$	$S^3 \times S^2$	5
2	$\mathfrak{g}_{12} = \text{span}(A_1, A_2)$	$U(1) \times U(1)$	$S^2 \times S^2$	4

The floors follow from two independent Hopf fibration calculations (Section 3.3): $U(1) \hookrightarrow S^3 \to S^2$ gives $\text{emb}(S^2) = 3$ and $\text{emb}(S^3) = 4$. The product rule gives the combined floors: $\text{emb}(S^2 \times S^3) = 3 + 4 - 1 = 5$ (not $3 + 4 = 7$, because the product can be embedded more efficiently than the naïve sum).

Closure verification: $[A_1, A_2] = 0 \in \text{span}(A_1, A_2)$ by R_{known}. The algebra is already closed. No extension needed.

Hyperedges: $\mathfrak{g}_0 \to \mathfrak{g}_1$, $\mathfrak{g}_0 \to \mathfrak{g}_2$, $\mathfrak{g}_1 \to \mathfrak{g}_{12}$, $\mathfrak{g}_2 \to \mathfrak{g}_{12}$. The lattice has two incomparable Tier 1 nodes — a branching point that the data must resolve.

What the pre-computed map certifies. Before seeing any data:

- Any quotient-faithful encoder requires $m \geq 4$ (the floor at $\mathfrak{g}_{12}$, the natural target quotient for two independent factors).
- An unconstrained disentanglement method [Locatello et al., 2019] that does not use the structural prior $[A_1, A_2] = 0$ cannot certify this floor in advance and cannot distinguish $\mathfrak{g}_1$, $\mathfrak{g}_2$, and $\mathfrak{g}_{12}$ without additional structure.
- The two Tier 1 nodes are incomparable: localising on $\mathfrak{g}_1$ vs $\mathfrak{g}_2$ requires the data to distinguish factor 1 from factor 2. If the task is symmetric in the two factors, the posterior $p(\mathfrak{g}' \mid \mathcal{D})$ may be bimodal at Tier 1.

Extension scenario: discovered coupling. Suppose after running Phases I–IV the posterior concentrates on $\mathfrak{g}_{12}$ but $\mathcal{L}_{\text{equiv}}$ retains a structured residual. Inspecting $\delta E(x)$ reveals a pattern that is not explained by $\exp(\alpha_1 A_1 + \alpha_2 A_2)$ — the residual has a component that varies with the *product* $\alpha_1 \alpha_2$, indicating a coupling between the two factors.

This is Case A: a missing generator A_{12} whose action is approximately $\rho(A_{12}) \approx A_1A_2 - A_2A_1/2$ (a second-order interaction). The residual regression identifies $\hat{A}_{12}$, and the extension $S \leftarrow S \cup \{\hat{A}_{12}\}$ triggers a recomputation of $\mathcal{H}$. The closure test then checks whether $[A_1, \hat{A}_{12}]$ and $[A_2, \hat{A}_{12}]$ lie in the new span. If they do, the algebra closes into a three-dimensional subalgebra of $\mathfrak{su}(2)$, and the correct quotient is a more complex space with a higher embedding floor — the two factors are not actually independent.

This example shows the extension procedure naturally discovering that the structural prior $[A_1, A_2] = 0$ was incorrect, and correcting it from the data.

9.4 The Four-Phase GCD Loop

The four phases implement the localisation of Section 9.2 on the pre-computed $\mathcal{H}(G_{\mathrm{spec}})$. The scheme is stratified: each phase produces a testable output gating entry to the next. It is not a convergence-certified algorithm; Phase III produces a conditional certificate, not a proof.

Phase I: Differentiable optimisation. Train the encoder E_θ and generator matrices $\mathcal{M}$ jointly by minimising the combined loss

$$\mathcal{L} = \mathcal{L}_{\mathrm{task}} + \mu\,\mathcal{L}_{\mathrm{equiv}} + \lambda\,\mathcal{L}_{\mathrm{closure}} + \nu\,\mathcal{L}_{\mathrm{radial}},$$

where $\mathcal{L}_{\mathrm{task}}$ is a downstream task loss, $\mathcal{L}_{\mathrm{equiv}}$ is the equivariance loss of Definition 8.2, $\mathcal{L}_{\mathrm{closure}}$ is the closure loss of Definition 8.5, $\mathcal{L}_{\mathrm{radial}}$ is the radial regulariser of Remark 8.3 (with $\nu = 0$ recovering the unregularised variant), and $\mu, \lambda, \nu \geq 0$ are weighting hyperparameters.

Output: trained pair $(\theta^*, \mathcal{M}^*)$.

Phase II: Algebraic closure certification. Evaluate $\mathcal{L}_{\mathrm{closure}}(\mathcal{M}^*)$. If $\mathcal{L}_{\mathrm{closure}} \leq \varepsilon_{\mathrm{closure}}$, declare $\mathcal{M}^*$ approximately closed and locate it as a node in $\mathcal{H}(G_{\mathrm{spec}})$. Otherwise diagnose the failure mode and proceed to Section 9.5.
Remark 9.2 (Sufficiency gap). $\mathcal{L}_{\mathrm{closure}} \leq \varepsilon$ is a necessary but not sufficient condition for $\mathcal{M}^*$ to span a genuine Lie algebra. At finite sample size, it is possible for $\mathcal{M}^*$ to pass the threshold while the posterior is concentrated at the wrong node of $\mathcal{H}(G_{\mathrm{spec}})$; see Open Problem 10.

Phase III: Topology certification. Match the closed $\mathcal{M}^*$ to its node in $\mathcal{H}(G_{\mathrm{spec}})$ and read off the pre-computed floor m^*. If $m^* \leq m$, issue a conditional certificate: *the architecture is not obstructed by the identified quotient.* If $m^* > m$, proceed to Phase IV.
Remark 9.3 (Covering ambiguity in Phase III). If $\mathcal{M}^*$ is consistent with multiple covering groups, the floor is the maximum over admissible coverings. The discriminant $\exp(2\pi M_i) \approx I$ can be evaluated on $\mathcal{M}^*$ to attempt disambiguation (Corollary 2.10).

Phase IV: Architecture correction. Expand the encoder to latent dimension m^* and return to Phase I with the expanded architecture. The learned $\mathcal{M}^*$ serves as a warm start. This is a one-time correction conditioned on the Phase III certificate.

9.5 Hypergraph Extension

When Phase II fails — either $\mathcal{L}_{\mathrm{closure}}$ does not converge or the posterior does not concentrate on any node — the pre-computed map $\mathcal{H}(G_{\mathrm{spec}})$ does not contain the correct algebra. Two structurally distinct failure modes arise, each with a different extension procedure.

Case B: Closure failure (bracket escape). $\mathcal{L}_{\text{equiv}}$ is low but $\mathcal{L}_{\text{closure}}$ refuses to vanish. The learned matrices individually explain the data but their brackets escape $\text{span}(\mathcal{M}^*)$. Define the *closure residuals*

$$R_{ij} = [M_i^*, M_j^*] - P_{\mathcal{M}^*}([M_i^*, M_j^*]), \qquad \|R_{ij}\| > \varepsilon_{\text{closure}}.$$

Each non-zero R_{ij} points to a direction in $\mathfrak{gl}(d)$ that the algebra is trying to close into. The *bracket completion* extension is:

$$S \;\leftarrow\; S \cup \big\{R_{ij}/\|R_{ij}\| : \|R_{ij}\| > \varepsilon_{\text{closure}}\big\}.$$

This is not a heuristic: $R_{ij}/\|R_{ij}\|$ is the unique direction in $\mathfrak{gl}(d)$ required to close the algebra at the current pair (i, j). Adding it to S and recomputing $\mathcal{H}$ is the minimal extension consistent with the data.

Termination: bracket completion terminates in at most $d^2 - \dim \mathfrak{g}_{\text{initial}}$ steps, since each extension adds at least one dimension to $\mathfrak{g}$ and $\dim \mathfrak{gl}(d) = d^2$ is finite.

Case A: Equivariance failure (missing generator). Both $\mathcal{L}_{\text{equiv}}$ and $\mathcal{L}_{\text{closure}}$ fail to converge, and the equivariance residual

$$\delta E(x) = E_\theta(T_\alpha(x)) - \exp\!\big(\textstyle\sum_i \alpha_i M_i^*\big) E_\theta(x)$$

has a *structured* component that is not isotropic. The structure indicates a missing generator A^* whose action on the latent space is $\rho(A^*)$. At small α, the residual is approximately $\alpha^* \rho(A^*) E_\theta(x) + \text{noise}$, so $\rho(A^*)$ is recoverable by the *residual regression*:

$$\hat{A} = \arg \min_{A \in \mathfrak{gl}(d),\, \|A\|=1} \sum_{x \in \mathcal{D}} \big\|\delta E(x) - \alpha^* A \cdot E_\theta(x)\big\|^2.$$

If the residual is genuinely structured (not isotropic noise), this regression has a meaningful solution. The extension is:

$$S \;\leftarrow\; S \cup \{\hat{A}\}.$$

Termination: not guaranteed in general. If the true group G^* has $\dim G^* < \infty$ (always the case for physical systems), the process terminates in at most $\dim G^* - \dim G$ extensions. If δE is isotropic — consistent with noise rather than a missing generator — the extension is flagged as inconclusive and no generator is added.

Remark 9.4 (Identifiability of the residual regression). The regression for $\hat{A}$ has a unique solution only if the data distribution $p_{\mathcal{D}}$ is rich enough to distinguish different generator actions on the latent space. If two matrices A and A' produce indistinguishable residuals on finite $\mathcal{D}$, the regression is ill-posed. This is the Case A identifiability problem; see Open Problem 10.

The alternating outer loop. The full extension procedure alternates between algebraic and statistical steps:

1. **Algebraic**: Compute $\mathcal{H}(G_{\text{spec}})$ from current S.
2. **Statistical**: Run Phases I–IV. Converge to $(\theta^*, \mathcal{M}^*)$.
3. **Diagnose**: Evaluate residuals.
   - If $\mathcal{L}_{\text{equiv}} \le \varepsilon$ and $\mathcal{L}_{\text{closure}} \le \varepsilon$: done.

- If $\mathcal{L}_{\text{closure}} > \varepsilon$: Case B. Extend S by bracket completion. Go to 1.
- If $\mathcal{L}_{\text{equiv}} > \varepsilon$ with structured residual: Case A. Extend S by residual regression. Go to 1.
- If $\mathcal{L}_{\text{equiv}} > \varepsilon$ with isotropic residual: inconclusive. Flag and stop.

This structure is analogous to the EM algorithm: the algebraic step (constructing $\mathcal{H}$) corresponds to the E-step (computing the expected sufficient statistics), and the statistical step (localising on $\mathcal{H}$) corresponds to the M-step (maximising the likelihood). Whether a formal EM convergence guarantee applies is Open Problem 10.

9.6 Regime-Dependent Initialisation

The two physical regimes of Section 4.4 determine both the hypergraph pre-computation cost and the initialisation of Phase I.

In **Regime A** (physics known), the hypergraph pre-computation is trivial: the Noether-derived catalog supplies the nodes and floor labels directly, and $|\mathcal{H}_{\text{prior}}|$ is small. Phase I is initialised from the catalog; Phase II is expected to pass immediately; Phase III reduces to a floor lookup. The extension procedure is unlikely to be needed.

In **Regime B** (physics unknown), the hypergraph pre-computation is the main cost: all subalgebras of the initial $\mathfrak{g}$ must be enumerated and labelled. Phase I must discover both the encoder and the generators simultaneously. TDA pre-screening [Moor et al., 2020, Dang-Nhu et al., 2026] estimates the topology of the data manifold and constrains the initial S before Phase I begins. Extension (Case A or Case B) is more likely to be needed.

Domain priors (fractal structure, energy cascade, sequence-driven folding) enter here as structured priors on $p(\mathfrak{g}' \mid \mathcal{D})$, reducing $|\mathcal{H}_{\text{prior}}|$ and improving the $O(\log |\mathcal{H}_{\text{prior}}|)$ sample complexity bound.

9.7 Trap Resistance

The GCD scheme provides counter-pressure against the three failure modes of Paper I, but does not provide mathematical guarantees of avoidance.

Traversal ambiguity. $\mathcal{L}_{\text{equiv}}$ forces consistency across orbits. The degenerate zero solution $E_\theta \equiv 0$ satisfies $\mathcal{L}_{\text{equiv}} = 0$ and $\mathcal{L}_{\text{closure}} = 0$ while carrying no topological content; $\mathcal{L}_{\text{radial}}$ (Remark 8.3) provides counter-pressure against this degenerate solution.

Task-map collapse. $\mathcal{L}_{\text{closure}}$ penalises representations whose generator set does not span a Lie algebra. A collapsed representation will typically fail the closure test; however, if the task loss dominates ($\lambda \to 0$), closure is not enforced and collapse remains possible.

Dimension trap. The pre-computed floor $m^*(\mathfrak{g}')$ at the identified node of $\mathcal{H}$ is the primary defence. If $m^* > m$, Phase IV corrects the architecture before task training proceeds. The defence is limited to quotients whose node is present in $\mathcal{H}$; missing generators (Case A) may carry a higher floor that is not discovered until after extension.

These are counter-pressures, not proofs. A fully certified scheme requires solution of Open Problems 10, 10, and 10.

9.8 Relation to Existing Architectures

The GCD scheme subsumes several existing approaches as special cases.

Group-equivariant CNNs [Cohen and Welling, 2016], homomorphism-equivariant encoders [Keurti et al., 2023], and physics-specialised architectures such as LorentzNet [Gong et al., 2022] fix G before training and enforce equivariance by construction. This corresponds to GCD with a fixed, known $\mathcal{M}$ (Regime A), skipping Phase II, with Phase III trivially satisfied. LorentzNet is an especially clear instance: the generator catalog is $\mathrm{SO}(1,3)^+$, derived from spacetime symmetry via the Noether chain, and the covering constraint is handled by restricting to the identity component. The GCD contribution is the discovery pathway for when G is not known in advance.

Disentangled representation learning methods [Locatello et al., 2019] correspond to Regime B without Phase II or Phase III: the topology of the latent space is unconstrained and neither the graded compression nor the embedding floor is addressed.

Symmetry discovery methods [Dang-Nhu et al., 2026, Ko et al., 2024] learn generators from data without the Serre sequence machinery. They correspond to GCD Phases I–II without Phase III, and without the hypergraph pre-computation or extension procedure. The pre-computed map, the floor certification, and the principled extension procedure are the contributions of the present paper beyond those methods.

10 Open Problems

Open Problem (Sufficiency of the commutator loss).

$\mathcal{L}_{\text{closure}}(\mathcal{M}) = 0$ is necessary for algebraic consistency of the learned generator family but not sufficient for quotient faithfulness. In terms of the pre-computed hypergraph $\mathcal{H}(G_{\text{spec}})$, the question is whether $\mathcal{L}_{\text{closure}} \leq \varepsilon$ correctly localises the posterior at the right node, or whether it can pass at the wrong node at finite sample size. Whether $\mathcal{L}_{\text{closure}} + \mathcal{L}_{\text{equiv}} + W_2(D_{\mathcal{X}}, D_Z)$ together suffice for quotient faithfulness under suitable conditions is open.

Open Problem (Sample complexity of generator recovery).

Given n transition pairs (x, x') from a system with true symmetry group G^*, how many samples are required to correctly localise on the node $\mathfrak{g}^* \in \mathcal{H}(G_{\text{spec}})$? By the finite hypothesis class bound, $O(\log |\mathcal{H}_{\text{prior}}|)$ samples suffice to distinguish nodes; the question is whether the equivariance loss provides a sufficiently accurate likelihood proxy to achieve this bound, and what the sample complexity is for the extension procedure when $\mathfrak{g}^* \notin \mathcal{H}(G_{\text{spec}})$.

Open Problem (Identifiability of the residual regression).

In Case A extension (Section 9.5), the residual regression for the missing generator $\hat{A}$ has a unique solution only if the data distribution is rich enough to distinguish different generator actions on the latent space. Formally: under what conditions on $p_{\mathcal{X}}$ and the encoder E_θ is the map $A \mapsto \sum_{x \in \mathcal{D}} \|\delta E(x) - \alpha^* A \cdot E_\theta(x)\|^2$ strictly convex on $\{A \in \mathfrak{gl}(d) : \|A\| = 1, A \perp \mathfrak{g}\}$? The connection to the transversal identifiability results of Khalil [2026] and to sparse recovery (when $\rho(A^*)$ is sparse in a known basis) should be investigated.

Open Problem (Convergence of the alternating extension procedure).

The outer loop of Section 9.5 — alternate between pre-computing $\mathcal{H}(S)$ and statistically localising on it, extending S from residuals when localisation fails — is structurally analogous to the EM algorithm. Does it converge to the true generator set S^* under the data-generating process? A sufficient condition likely requires: (i) identifiability of $\hat{A}$ at each extension step (Open Problem 10), (ii) the true group G^* has finite dimension, and (iii) the data distribution is not concentrated on a measure-zero subset of orbits. The pathological cases are: multiple incomparable branches in $\mathcal{H}$ at finite sample size, and Case A extensions that are consistent with the data but not with G^* (spurious generators).

Open Problem (Covering type from data).

Given a learned closed algebra $\mathfrak{g}$ and transition data, can the covering type be determined at finite sample size? The discriminant $\exp(2\pi A_i) \approx I$ can be checked if large-angle transitions are observed; the sample complexity of observing such transitions with sufficient coverage is unknown in general.

Open Problem (Topological bounds for formal type quotients).

The Serre long exact sequence and the embedding floors of Paper 3 apply when there is a continuous group action on a topological space. For quotient types in formal mathematics, the "space" is the type A and the "group" is the groupoid generated by the relation R. Whether the classifying space $B\langle S \mid R\rangle$ provides useful topological bounds on $\mathrm{emb}(A/R)$ is open. The Spivak lifting-constraint framework [Spivak, 2014] provides the categorical language; the topological content requires the additional structure of a geometric realisation of the type quotient.

Open Problem (Bayesian generator discovery: posterior concentration).

Under what conditions does the posterior $p(\mathfrak{g}' \mid \mathcal{D})$ of equation (8) concentrate on the true node $\mathfrak{g}^* \in \mathcal{H}(G_{\text{spec}})$ as $|\mathcal{D}| \to \infty$? The covering ambiguity means the posterior can concentrate only up to the finite set of admissible coverings; identifying which covering is selected requires the discriminant data of Open Problem 10. A formal consistency result for the posterior would parallel the identifiability results of Dang-Nhu et al. [2026].

A Definitions and Key Concepts

This appendix collects the principal formal definitions from the main text in one place. Concepts are ordered by first appearance. All concepts from the generator framework are illustrated with the running example of Section 3.3: a $U(1)$ generator acting freely on S^3 via the Hopf fibration, with quotient $Q = S^2$.

Generator specification G_{spec}. A 4-tuple $(S, R_{\text{known}}, \tau, \mathcal{E})$ where S is a finite set of generator symbols, R_{known} is a set of known relations, $\tau \in \{\text{Lie}, \text{discrete}\}$ is a type tag, and $\mathcal{E}$ is an observation space (Definition 2.1). In the running example: $S = \{A\}$ (one $U(1)$ generator), $R_{\text{known}} = \emptyset$, $\tau = \text{Lie}$, $\mathcal{E} = S^3$.

Admissible quotient class $\mathcal{Q}(G_{\text{spec}}, \mathcal{X})$. The set of quotient spaces Q (up to homotopy equivalence) consistent with G_{spec} acting on $\mathcal{X}$ under the regularity hypotheses H1–H3 (Definition 2.3). In the running example: $\mathcal{Q} = \{S^2\}$ (unique, since the Serre sequence forces $\pi_2(Q) \cong \mathbb{Z}$ and $Q \simeq S^2$).

Regularity hypotheses H1–H3. H1: the generator set S spans a Lie algebra $\mathfrak{g}$ (algebraic closure). H2: the induced group action on $\mathcal{X}$ is free and proper. H3: $\mathcal{X}$ is contractible (Definition 2.6).

Algebraic closure. S is algebraically closed if $[M_i, M_j] \in \text{span}(S)$ for all $M_i, M_j \in S$, where $[\cdot, \cdot]$ is the matrix commutator (Definition 3.1).

Covering ambiguity. Two Lie groups G_1, G_2 with the same Lie algebra $\mathfrak{g}$ (e.g. SO(3) and SU(2)) cannot be distinguished from $\mathfrak{g}$ alone; they are related by a covering map (Corollary 2.10).

Architecture obstruction. If all $Q \in \mathcal{Q}(G_{\text{spec}}, \mathcal{X})$ satisfy $\text{emb}(Q) > m$, no quotient-faithful encoder $E_\theta : \mathcal{X} \to \mathbb{R}^m$ exists (Proposition 2.11).

Generator bottleneck. The constraint $E_\theta(T_\alpha(x)) = \exp(\sum_i \alpha_i M_i)\, E_\theta(x)$ for all $x \in \mathcal{X}$ and α, enforcing equivariance of the encoder with respect to the parametrised group action (Definition 8.1).

Equivariance loss $\mathcal{L}_{\text{equiv}}$. Mean squared deviation from the generator bottleneck over the training sample and a distribution $\mathcal{A}$ over group parameters (Definition 8.2).

Closure loss $\mathcal{L}_{\text{closure}}$. Sum of squared residuals $\|[M_i, M_j] - P_{\mathcal{M}}([M_i, M_j])\|_F^2$ measuring failure of the learned generator matrices to span a Lie algebra (Definition 8.5).

Radial regulariser $\mathcal{L}_{\text{radial}}$. Negative log-likelihood of the radial marginal of a D-dimensional Gaussian, preventing representational collapse to the origin (Remark 8.3).

B Proofs and Proof Sketches

B.1 Proof of Proposition 3.4 (Algebraic closure determines $\mathfrak{g}$)

Let $\mathcal{M} = \{M_1, \ldots, M_k\}$ be a finite set of matrices in $\mathfrak{gl}(d)$ satisfying $\mathcal{L}_{\text{closure}} = 0$, i.e. $[M_i, M_j] \in \text{span}(\mathcal{M})$ for all i, j. Then $\mathfrak{g} := \text{span}(\mathcal{M})$ is closed under the Lie bracket by assumption, and under scalar multiplication and addition by linearity. Thus $\mathfrak{g}$ is a Lie subalgebra of $\mathfrak{gl}(d)$. By Ado's theorem, every finite-dimensional Lie algebra over $\mathbb{R}$ has a faithful finite-dimensional representation, so $\mathfrak{g}$ is the Lie algebra of a connected Lie subgroup $G \leq GL(d)$ by the closed subgroup theorem (Lee [2013], Theorem 20.12). The group G is determined up to isomorphism by $\mathfrak{g}$ together with a choice of covering; the covering ambiguity is characterised in Corollary 2.10. $\square$

B.2 Comparison with Paper 3 embedding floors

Paper 3 [Farkas, 2026c] derives embedding dimension lower bounds via characteristic classes and the Stiefel–Whitney and Pontryagin obstructions. The floor $\mathrm{emb}(Q) \geq m^*$ in Proposition 8.7 of the present paper is of the same type but derived via the Serre long exact sequence rather than characteristic class arguments. The two derivations are complementary: the Serre sequence gives a bound from the homotopy groups of Q, while the characteristic class method gives a bound from cohomological obstructions. For the Hopf fibration example, both give $\mathrm{emb}(S^2) \geq 3$.

C Contribution Attribution and AI Disclosure

C.1 Context

Paper 6 was developed across multiple working sessions in March–April 2026. The primary drafting tool was Claude Sonnet 4.6 (Anthropic). Separate editorial review and calibration feedback sessions were conducted using ChatGPT 5.4 (OpenAI) and Gemini 3 Pro (Google), with resulting recommendations integrated by the first author.

The central conceptual contribution — the generator specification as a mediating object between partial structural knowledge and quotient constraints, and the two-step chain connecting it to an embedding floor — was the author's framing. The Hopf fibration was selected as the primary worked example through directed sessions that rejected SO(3) (algebra closes trivially under the commutator test, so no meaningful example) and settled on $U(1)$ acting on S^3 because the Serre sequence collapses cleanly and the bound is sharp.

The Noether-to-catalog chain (conservation laws → conserved charges → Lie group families) and the Bayesian model selection framing for covering ambiguity in Regime A were the author's contributions; the AI role was execution and verification. The identification of algebraic closure as a differentiable, certifiable condition (rather than a static architectural property), and the honest statement of the sufficiency gap, were developed jointly.

The degenerate zero solution ($E_\theta \equiv 0$ satisfying both $\mathcal{L}_{\text{equiv}} = 0$ and $\mathcal{L}_{\text{closure}} = 0$) was identified by the AI as an unsolicited structural observation during review of the GCD scheme; the author directed the integration and scope statement. The connection to the HALO radial regulariser was also AI-identified. The curation of Spivak citations — specifically the rejection of overclaimed categorical justifications for the GCD training loop and the identification of which connections are mathematical identities vs analogies — involved active correction by the author against AI-tool overclaiming.

The generator hypergraph framing — pre-computing the complete search space from the generator specification before any data is seen, and separating the algebraic and statistical computations — emerged from an extended dialogue in which the author proposed the key conceptual separation; the AI contributed the lattice/tier formalisation. The Case A residual regression for missing generators and the alternating outer loop (EM analogy) emerged as joint contributions in which the author identified the structural gap (the map may be incomplete) and the AI proposed the regression formulation and convergence analogy. Both are marked AI ++ as unsolicited structural suggestions that changed the direction of the section.

C.2 Contribution Weight by Result

Columns: **Lit.** = prior literature; **Author** = Peter Farkas; **AI** = the AI tools used in drafting this paper (Claude Sonnet 4.6 / Anthropic, ChatGPT 5.4 / OpenAI, Gemini 3 Pro / Google, MetaAI / Meta). Weight key: **0** = not applicable; + = substantive contribution; ++ = primary responsibility (this party could have produced this result independently). The AI column records the maximum weight across AI tools for a given row. AI ++ is reserved for results that emerged as unsolicited structural contributions by an AI tool, not execution of a direction.

Result / contribution	**Character**	**Lit.**	**Author**	**AI**
Generator specification $G_{\text{spec}} = (S, R_{\text{known}}, \tau, \mathcal{E})$ as the mediating object between partial structural knowledge and quotient constraints	conceptual invention	0	++	+
Admissible quotient class $\mathcal{Q}(G_{\text{spec}}, \mathcal{X})$: formalising which quotients are consistent with a specification without naming $\sim$	formalisation	0	++	+
Two-step chain: algebraic closure (horizontal) then Serre LES (vertical) as the route from generator spec to embedding floor	synthesis	+	++	+
Algebraic closure as a *certifiable, differentiable* condition ($\mathcal{L}_{\text{closure}}$) rather than a static architectural property, with honest identification of necessary-not-sufficient status	operationalisa	0	++	+
Covering ambiguity identified as a *finite* model-selection problem: the catalog of admissible coverings is small and discriminable via $\exp(2\pi M_i) \approx I$	identification	++	++	+
Hopf fibration as primary worked example: rejection of SO(3) (algebra closes trivially) in favour of $U(1)$ on S^3 where the Serre sequence collapses cleanly and the bound $\text{emb}(S^2) \geq 3$ is sharp	selection + judgement	+	++	+
Noether-to-catalog chain (Eq. 7): conservation laws $\to$ conserved charges $\to$ Lie group families as a *principled* generator dictionary for Regime A, making catalog generation non-ad-hoc	synthesis	+	++	+

Result / contribution	Character	Lit.	Author	AI
Bayesian model selection over the Noether-derived catalog as the principled resolution of covering ambiguity in Regime A, framing the SO(3)/SU(2) choice as a posterior inference problem, not a lookup	framing	0	++	+
Regime A/B distinction and its consequences for GCD initialisation: physics-known regime starts from catalog; physics-unknown regime requires TDA pre-screening before any group structure is assumed	design	0	++	+
GCD Phase III as a *pre-training certificate*: the topology floor check is not a convergence test but a constraint on whether training can succeed at all, prior to optimisation	framing	0	++	+
GCD four-phase stratified structure with explicit phase gates and honest non-claims (scheme, not convergence-certified algorithm)	design	0	++	+
BDF Consistency Hyperedge identified as a lifting constraint in Spivak's sense: mathematical identity, not analogy; extends admissibility framework to formal theorem provers	identification	+	++	+
$E_\theta \equiv 0$ as a degenerate global minimiser of $\mathcal{L}_{\text{equiv}} + \mathcal{L}_{\text{closure}}$ carrying no topological content (traversal ambiguity in concrete form)	identification	0	+	++
Connection of HALO radial NLL [Pisoni, 2026] to the degenerate zero problem; $\mathcal{L}_{\text{radial}}$ as counter-pressure	cross-reference	+	+	++
Identification of which Spivak connections are mathematical identities vs analogies, and rejection of overclaimed categorical justifications for GCD	curation + rejection	0	++	+
Generator hypergraph $\mathcal{H}(G_{\text{spec}})$: precomputation of the complete search space from G_{spec} alone, before data; tier structure and floor labelling	conceptual invention	0	++	+
Case B bracket completion: extracting R_{ij} as principled extension direction when $\mathcal{L}_{\text{closure}}$ fails to vanish	operationalisation	0	++	+

Result / contribution	Character	Lit.	Author	AI
Case A residual regression: recovering missing generator action from structured equivariance residual; sparse recovery connection	invention	0	++	++
Alternating outer loop (EM analogy): algebraic pre-computation and statistical localisation as decoupled alternating steps 0	synthesis	0	++	++

C.3 Attribution Prompt for AI-Assisted Papers

The following prompt was used to calibrate attribution in this series. It can be adapted for any AI-assisted academic paper.

> *You are helping audit the intellectual contributions of an AI-assisted academic paper. For each result, theorem, definition, or synthesis, assign contribution weights: ++ (primary responsibility — this party could have produced this result independently), + (substantive contribution),* ***0*** *(not applicable). Be honest. If the author provided the key insight and the AI only executed it, the author gets ++. If an unsolicited structural suggestion from the AI changed the direction of a proof or section, the AI gets ++. Record the results in a table with columns: Result, Character, Literature, Author, AI.*

Notation and Key Concepts

This section gives plain-language descriptions of the key concepts used throughout the paper. Readers familiar with Lie theory and algebraic topology may skip it. Concepts are grouped thematically and illustrated with the Hopf fibration running example: a $U(1)$ generator acting freely on S^3 with quotient S^2.

Generator specification. A partial structural description of the data-generating process. Instead of declaring the equivalence relation $\sim$ fully (Route A), the practitioner specifies a finite set of infinitesimal symmetries and any known relations among them. In the running example: one generator A with $[A, A] = 0$ (trivially satisfied for a one-dimensional algebra), acting on S^3.

Admissible quotient class. The set of all quotient spaces consistent with the generator specification. Constraints flow from generator algebra to group topology to quotient topology via the Serre long exact sequence. In the running example: the unique admissible quotient is S^2.

Serre long exact sequence. A tool from algebraic topology that relates the homotopy groups of a total space, a fibre, and a base. For a principal G-bundle $G \hookrightarrow \mathcal{X} \to Q$ with $\mathcal{X}$ contractible, it gives $\pi_k(Q) \cong \pi_{k-1}(G)$ for $k \geq 2$ and $\pi_1(Q) \cong G$. This is the "vertical step" that converts generator information into quotient topology.

Embedding dimension floor $\mathrm{emb}(Q) \geq m^*$**.** A lower bound on the dimension of any Euclidean space into which Q can be continuously and injectively embedded. If the latent space dimension $m < m^*$, no quotient-faithful encoder exists. In the running example: $\mathrm{emb}(S^2) \geq 3$.

Algebraic closure. The property that the Lie bracket (commutator) of any two generators lies in the span of the generator set. Necessary for the generators to span a valid Lie algebra. Measured by the closure loss $\mathcal{L}_{\mathrm{closure}}$.

Covering ambiguity. Two groups with the same Lie algebra (e.g. SO(3) and SU(2)) cannot be distinguished from generator matrices alone. Disambiguated by checking whether $\exp(2\pi M_i) \approx I$ for each generator.

GCD scheme. A four-phase training and certification procedure: (I) joint optimisation of encoder and generator matrices; (II) algebraic closure certification; (III) topology library match and dimension floor check; (IV) architecture correction if $m < m^*$. Each phase produces a testable output gating entry to the next.

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